EIGSEP Memo 008

Marjum Pass 2026-07: image-domain modeling of the sky, the ground and the Sun

EIGSEP Team

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Abstract

How calibrated Marjum spectra are modelled in the image domain: as the beam-weighted sum of a Galactic sky map, a terrain at one uniform temperature below the DEM horizon, a receiver offset, and the Sun. The memo covers the model and the degeneracies its geometry leaves, the night fit to the calibrated 07-17/18 data, and the Sun. Box-air records solar radio bursts at the times independent spectrographs recorded them, which identifies solar emission in the data. The quiet Sun is harder: the ridges hide it for hours, so it appears only at a handful of terrain crossings. The one calibrated crossing, 15:54 UTC on 07-17, shows an excess 3.6 times the concurrent Sun's prediction (1.4 scaled by the template amplitude), while the raw data at the other crossings allow a Sun of the expected size but do not detect it convincingly. Each measurement is shown against the raw data around it. Finally, regressions on the rotating antenna give a repeatable ground term, too frequency-dependent to be the ground's temperature alone. Over phase C, raw power at 62–82 MHz improves with a slow additive daily term, but skies shifted in sidereal time do nearly as well, so it is not shown to be sky. With a gain free to change within the hour, as the model requires, a phase-C ground term follows at 110–205 MHz, with errors from injected mocks; at the frequencies where the fit collapses it is not determined, and the fast gain follows neither a recorded stream nor the receiver's measured gain. The antenna's heights follow the rebuilt mode table: the 07-15 night and the 07-17 sunset were on the ground, and are left out until a horizon profile at 2 m exists. Every number, figure and table is computed by `make_products.py` in this directory from the products under `derived/ground_sky/` (Appendix A). The kelvin scale is memo 004's, the beam is HFSS (memo 005 measures it), and pointing is memo 003's.

1 Summary

- **The model** is linear at each frequency: sky pixels, one ground temperature, one offset per receiver regime and, optionally, the Sun's flux density. The columns are integrated through an assumed beam on the real geometry (Section 2). A uniform shift of sky and ground against the offset is an exact null direction.
- **The DEM stops short of the southern horizon.** Extending it 4 km south raises the horizon by up to 5.47° at 87.5 m and changes the predicted antenna temperature by up to 47 K, with the sky and beam fixed (Section 2, Fig. 1). The extended DEM and its horizon are now `derived/dem/v0004` and `horizon_profiles_v0004`, and every fit and product here now uses them.
- **Heights.** The mode table rebuilt on 2026-10-18 (memo 001) puts the 07-15 night and the 07-17 sunset on the ground, not at 30 m, and part of the 07-18 night at unknown height. Box-air's lidar and IMU agree wherever they can test it (§4.1). Every fit here uses only rows at 30, 87.5 or 91 m; the ground stretches wait for a horizon profile at 2 m.
- **The ground temperature is not measured by the calibrated data.** They hold one pointing, one height and one regime at night (165 two-minute bins), where $T_{\rm gnd}$ and the offset are degenerate. Only their sum, the pedestal, is determined: 12 to 90 K at 110–170 MHz. A forecast on the full campaign geometry shows that height changes are what would break the degeneracy (Table 1).
- **The data vary about twice as much as GSM through HFSS predicts**: the template amplitude is $a \approx 2.1$ (1.3–3.1 over the band). The model also does not reach the noise: median χ^2 per bin is 120, and 234 on held-out blocks.
- **When the Sun is visible.** The ridges stand 39° high at 352 m to the east and 17° at 171 m to the west. The Sun clears the ridge at **15:54 UTC on 07-17** (87.5 m), 3.5 h after it reaches 0° altitude, and sets

behind the ridge at **01:21 on 07-18** (91 m). The calibrated data therefore see the direct Sun only from 15:54 to 16:22 on 07-17. At these wavelengths the edge diffracts, so the Sun fades in over 10–30 min.

- **Box-air sees the Sun's radio bursts.** At the times e-CALLISTO spectrographs recorded type III bursts, raw box-air shows emission drifting from high to low frequency within seconds, bright at 45–85 MHz as those stations saw it. 5 of 12 bursts with data pass a test that 5.2% of random daytime times pass ($p = 2.2 \times 10^{-4}$). That identifies solar emission in our data physically (§6.5).
- **The quiet Sun across days.** All 4 terrain crossings of phase C are shown in their raw context; 3 of them, on three days and three heights, can be fitted. They allow a Sun term of 3–16 SFU at 121–219 MHz, the size of the concurrent Sun (RSTN at 245 MHz, steady at 13–17 SFU from 07-15 to 07-18). They do not detect it convincingly, since raw power drifts by as much for other reasons (§6.7). None shows a jump as large as the calibrated 07-17 one.
- **The calibrated 07-17 sunrise excess is larger than the Sun.** After 15:54 the calibrated spectrum rises by a smooth, additive 20–25 K at 148–199 MHz (Fig. 6). That is 3.6 times what concurrent RSTN flux predicts (1.4 times scaled by a), and 2.0–3.2 times the other crossings at 199 MHz. A local fit gives $S_\odot$ from 7 SFU at 109 MHz to 32 SFU at 199 MHz; the same fit at the 0° sunrise, with the Sun behind the ridge, gives −6 to 1 SFU. Part of the excess is probably not the Sun. Something that changes as the box comes into sunlight would follow the same geometry.
- **Impulsive events in daytime**: holding the height era, the local hour or the Sun's height above the horizon fixed in turn, the controls disagree with each other and between eras, so whether the events follow the Sun is open (§6.6).
- **Not yet possible**: separating the Sun from the lit ground by timing, or from the sky and ground by the beam's sweep during the raster (§6.8). Below 90 MHz the calibrated result is unusable.
- **The ground from the rotating antenna.** When the antenna moves, the ground fraction swings while the gain stays put, so the ground/GSM ratio of a gain-free regression gives the ground's brightness in GSM kelvin. Three 91 m windows agree on it. It falls from 466 K at 125 MHz and 346 K at 139 MHz to 196–246 K at 150–175 MHz and reaches thousands below 85 MHz, which a reflected-sky term does not explain (§7). A joint fit across frequency and the sky-calibrated parked nights do not separate the ground, and low-order corrections to GSM are not separable from ground-fixed structure.
- **The low-frequency daily term over phase C.** On rows at confirmed heights, a slow additive term at 62–82 MHz still improves the fit with 3-h gain knots over a constant ground, but the evidence that it is sky has gone: skies shifted in sidereal time do nearly as well, it is absent from the second half of phase C and from 07-16 alone, and the fitted gain follows the open-sky mean even with 3-h knots. Earlier versions found it mostly in rows now known to be on the ground. Whether it is additive at all is open (§7.5).
- **The ground term with a fast gain.** Over phase C the model needs a gain that changes within the hour: with 3-h knots $T_{\rm gnd}$ comes out 1.7–4.5 times lower, and an injected mock with the 20-min gain refitted with 3-h knots reproduces those values to within 110 K. With 20-min knots, $T_{\rm gnd}$ falls from 425 K to 199 K over 110–205 MHz, with injection errors of 33–122 K, 0.9–2.6 times the fit's own. At 180, 188, 219 MHz the fit has two solutions the data cannot tell apart, one with the GSM amplitude collapsed, and no ground term is quoted; 51, 82, 117, 125, 172, 227, 234 MHz are unstable. The fast gain follows no recorded stream, the RF switch's temperature included; the low-band collapse and the temperature relation of earlier versions came with the rows on the ground. Where the receiver's gain after the RF switch is measured (07-17, 87.5 m) it moves by about 1.1% and does not follow the sky, while the fitted gain moves by up to 14.9%; so the fast gain is a gain change before the switch or the fit absorbing model mismatch. A fast gain shared within each band, fitted jointly to convergence, rescues 180, 188 MHz; shared across both bands, it changes the high-band ground term by up to 43%, so that choice is open (§7.6).
- **A solar burst at the 07-17 sunset.** Box-air alone shows a broadband excess below 100 MHz during a C1.5 flare (23:51–00:09 UTC), with the antenna on the ground and tilted, not at 30 m. What ends it is open: from the ground there is no traced horizon, so the earlier reading as the ridge occulting the Sun is withdrawn (§6.7).

2 The model

At each frequency, every time bin t is modelled as

$$T(t) = \sum_p A_{\rm sky}[t,p]\, T_{\rm sky}[p] + A_{\rm gnd}[t]\, T_{\rm gnd} + T_{\rm off}[r(t)] + A_\odot[t]\, S_\odot, \tag{1}$$

with the sky in Galactic HEALPix pixels ($N_{\rm side} = 8$), the ground everything below the DEM horizon at one temperature, an offset per receiver regime r, and the Sun's flux density $S_\odot$ in SFU ($10^{-22}\,\mathrm{W\,m^{-2}\,Hz^{-1}}$). The sky is written $T_{\rm sky} = a \cdot \mathrm{GSM} + \delta$. The amplitude a is free; it absorbs the front-end loss (calibrated temperatures are at reference plane P, memo 004) and any GSM scale error. Only δ has a prior, a Gaussian of 10% of GSM.

Integration. The sky and ground columns are integrated on a topocentric grid ($N_{\rm side} = 64$). For each distinct pointing and height era, the beam (HFSS total power here) is rotated by the mount, `mount_rotation`$(\mathrm{az} + \mathrm{az}_0, \mathrm{el} + \mathrm{el}_0, \psi)$, with $\mathrm{az}_0 = -16.04°$, $\mathrm{el}_0 = +0.93°$ and $\psi = 142.2°$ from the reference beam fit `empirical_raster_v0013` (memo 005), the frame in which its empirical beam was fitted. It is cut at that era's DEM horizon (`horizon_profiles_v0004`, traced from the antenna of geometry release `v0004`, memo 002, on the DEM mosaic `derived/dem/v0004`, which extends v0003 6 km south). Only the map from grid directions to Galactic pixels changes with time. Rows are normalized by the full-sphere beam integral, so sky and ground weights sum to one. That makes a uniform shift of sky and ground, against the offset, an exact null direction: breaking it needs a known offset or a known sky. The solver (`eigsep_sim.design_matrix.solve`) reports which columns move along a null direction and marks them unconstrained.

Bearings. The DEM, and so the horizon profiles, are in NAD83(2011)/UTM 12N. Grid north there is 1.52° from true north. The fit rotates the horizon to true bearings; the first trials of 2026-10-01 did not.

Terrain coverage. Earlier versions of this memo traced the horizon only out to the edge of the DEM mosaic `derived/dem/v0003`, which reaches $N = -1000$ m on its southern side. To see what that edge costs, we extended the mosaic 4 km south on the same grid, keeping every original pixel, the origin, the resolution and the vertical datum, and traced the horizon again from the profile antenna position $(E, N, U) = (1656.2, 2031.8, 1778.6)$ m (the 91 m era of geometry release `v0004`) in the DEM frame. The 30, 87.5 and 91 m height labels subtract 61, 3.5 and 0 m from U. Bearings are true bearings, rotated from grid north by 1.52° as above, in 0.1° steps with 720 sub-samples of the mask. Only the DEM changes between the two traces: sky, beam, site, times, pointings and frequencies are the same. The southern terrain raises the horizon by up to 3.45°, 5.47° and 5.51° at 30, 87.5 and 91 m. At the eight observed night pointings and four frequencies (51–246 MHz), the predicted antenna temperature changes by at most 16, 47 and 48 K (Fig. 1). The 87.5 m case is the selected night (07-17, 04:22–11:23 UTC); the 30 and 91 m cases are synthetic height controls evaluated at those same times, not observations from those eras. These are differences between two model predictions, not measured errors in any calibrated spectrum. A third trace with 8 km of added terrain differs from the 4 km one by at most 4.1×10^{-12} K, at the nominal position and six ±3 mm displacements. Edge flags alone do not test coverage: the original DEM raised none at 30 m and still changed. The tracing is also not shown to be accurate to sub-mK: the finest bearing refinement tested changes the prediction by up to 7.6 mK, so the numbers above are diagnostic and not tied to any convergence bound. This paragraph and Fig. 1 compare that original DEM with the extension; every fit and product in this memo now uses the extended DEM v0004 below.

The extended terrain is now a product. `derived/dem/v0004` is v0003 plus 6 km to the south, on the same grid and keeping every v0003 pixel, and `curation/horizon_profiles_v0004` is the horizon `build_horizon_profiles.py` traces on it. Over v0003 it rises by at most 3.45, 5.47 and 5.51° at 30, 87.5 and 91 m, on 209 bearings at 87.5 m, and is unchanged elsewhere. Terrain outside v0004 cannot change it: `coverage_certificate.py` follows every bearing beyond the DEM edge on the USGS 1 arc-second DEM, out to the distance (at most 42 km) beyond which terrain below 4200 m would sit under the skyline, and finds none that rises above it, from the profile position at all three heights or from the eight displaced positions of the profile's sensitivity block (`curation/horizon_profiles_v0004_coverage.json`). The coarse terrain is raised by 8.5 m, the most by which it falls below the skylines traced inside the DEM; the smallest margin is then 0.99°, on southern bearings, and 2.57° in the other directions. The check covers this antenna position only.

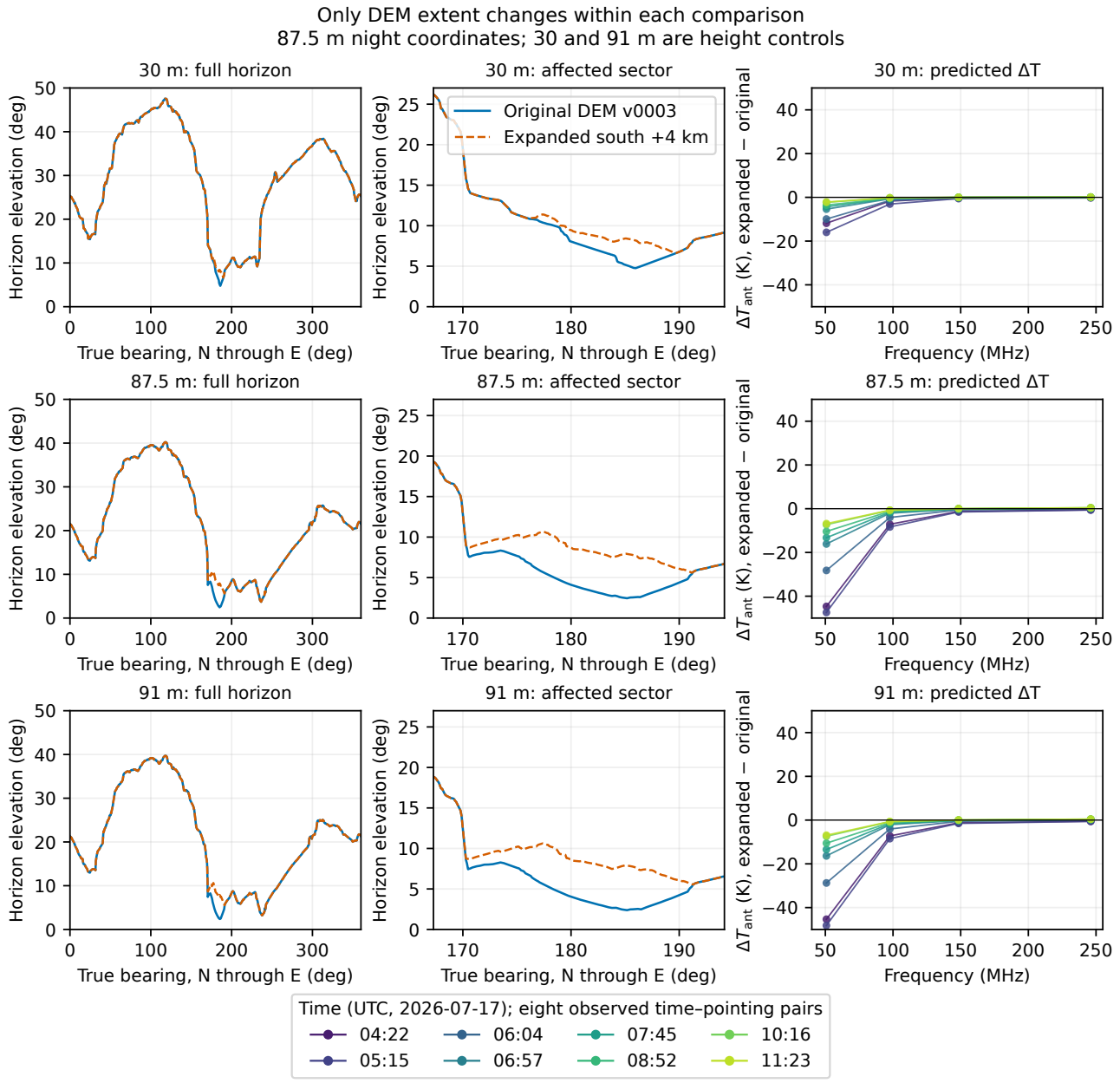


Figure 1: Effect of extending the DEM 4 km south, at the nominal antenna position, for the three height labels (rows). The horizon from the original DEM (`derived/dem/v0003`) and the extended DEM, with the affected southern sector enlarged, and the extended-minus-original predicted temperature at four frequencies, with the sky and beam fixed. Colours mark the eight observed night time/pointing pairs by UTC on 07-17; points are the evaluations and lines guide the eye. Within each pair only the DEM changes. The 30 and 91 m rows are synthetic height controls at the same times. The model azimuth is 0.865° for all eight pairs and the model elevations are 0.63–1.33° (`inputs/terrain_coverage/coordinates.csv`). Bearings are true; the horizon step is 0.1°.

The Sun column. The Sun is a point source:

$$A_\odot(t) = \frac{\lambda^2}{8\pi k}\, G(\hat{s}(t))\, V(t) \quad \text{(K per SFU)}, \tag{2}$$

for one linear polarization and an unpolarized source. Here G is the beam gain toward the Sun, with the same full-sphere normalization as the rows, and V is its terrain visibility. V is not read off the fixed horizon profile. The DEM is traced along the Sun's own bearing at each time, which gives the edge's elevation and its distance d. The ridges are 150–350 m away, so at these wavelengths the edge diffracts over $\sqrt{\lambda d/2} \approx 2$–$6°$, tens of minutes of solar motion. V is the Fresnel knife-edge power for the Sun's angle θ above the edge, $v = -\theta\sqrt{2d/\lambda}$: it is 0.25 at $\theta = 0$ and rises to 1 with lit-side ripples. The real ridge is rough and rounded, which smooths these ripples.

The knife-edge shape is an assumption. No measurement here tests it: its frequency-dependent onset, its value at the edge, its ripples. Every fit uses $A_\odot$ only as a template, with one fitted amplitude and, in the timing scans, one time shift. Figure 5 shows it against the data it was fitted to.

3 What the geometry can constrain

Before any data are used, `degeneracy_study.py` builds the design matrix on the campaign's real pointing and heights. It reports Fisher errors on $T_{\rm gnd}$ with radiometer noise in 120 s bins (Table 1).

- With the sky free, $T_{\rm gnd}$ is never constrained. That holds even with a known offset: within one pointing and era the ground column is constant and trades against a uniform sky shift.
- With a 10% GSM prior and the offset free, $T_{\rm gnd}$ and the offset are anticorrelated at $\rho = -0.95$ to -1.00. Height changes, which move the horizon, break the degeneracy: all of phase C would give 0.7–16 K.
- The calibrated windows alone are weak unless the offset is known.

The study also simulates and recovers. With HFSS as the true beam the recovered $T_{\rm gnd}$ is unbiased. With the empirical beam as truth and HFSS assumed, it is biased by up to 38 K (16σ) at 78 MHz, while χ^2 per row moves only from 5.5 to 5.6. A good χ^2 cannot vouch for $T_{\rm gnd}$.

slice	58.6	78.1	101.6	148.4	199.2
10% GSM sky prior, offset free					
parked, 30 m only	752.6	391.7	256.6	89.6	128.9
parked, all eras	19.7	9.3	5.0	2.4	0.9
all of phase C	16.3	7.7	3.9	1.6	0.7
tcal windows only	65.5	18.7	6.8	6.0	0.9
10% GSM sky prior, offset known to 1 K					
parked, 30 m only	8.2	3.9	2.2	2.0	1.8
parked, all eras	4.6	2.3	1.5	1.3	0.7
all of phase C	3.4	1.7	1.2	1.0	0.6
tcal windows only	8.4	3.4	1.7	1.8	0.7

Table 1: Forecast $\sigma(T_{\rm gnd})$ in K at five frequencies (MHz), from the degeneracy study (`summary.json` in `data-analysis`; radiometer noise, 120 s bins). "–": unconstrained. This forecast predates `tcal` v0003 and uses the v0000 calibrated windows.

4 Data

Box-air, antenna-switch rows calibrated by `tcal@v0003` (memo 004), which cover 07-17 04:11–16:22 (87.5 m, receiver regime rx-A) and 07-18 01:27–02:56 (91 m, rx-B). Samples flagged by `flags@v3` (memo 006) are dropped, on any bit but the advisory bit 9. From 07-17 15:36 the transmitter's comb channels (every eighth channel) are dropped whatever the flags say. At each HFSS beam frequency the remaining channels within ±2 are averaged. Rows are then averaged into static bins of one era, regime and pointing (az and el each within 1°): 302 two-minute bins (275 rx-A, 27 rx-B; product `binned_v0003`) and 615 one-minute bins (`binned_v0004`). Each bin's

noise is the standard error of its rows. Frequencies whose in-bin scatter exceeds ten times the radiometer expectation are not fitted, which removes FM and leaves 44.

4.1 Heights

Every fit here takes the antenna's height from the mode table as rebuilt on 2026-10-18 (memo 001, `derived/height_eras/v0001`), by source file, and uses only rows at a height with a horizon profile (30, 87.5 or 91 m). That table moves three stretches that earlier versions of this memo treated as at height. The 07-15 night and 07-16 23:27 to 07-17 01:15 UTC, called 30 m before, are on the ground (about 2 m). Around 07-18 00:00, called 91 m, the antenna was lowered and raised again, and its height is unknown. Of phase C's 17102 raw bins (`raw_binned_v0002`), 2812 are now at 30 m, 3009 on the ground and 1571 of unknown height; with the earlier labels (`raw_binned_v0001`, for comparison only), 6904 were at 30 m.

`validate_heights.py` checks the relabelled stretches against box-air's own recorded lidar range and IMU elevation, without the mode table, in all 3502 box-air files of 07-14 to 07-18 (Fig. 2, Table 2). The lidar returns a distance only in a few stretches; elsewhere every sample reports an error.

- **07-18, lowered and raised again.** From 07-17 23:30 the lidar reads a median of 1.1 m, the box on the ground. Over the next half hour it reads 31 m (28–39), going up, and then nothing within range.
- **The 07-17 sunset, on the ground.** From 07-16 23:27 to 07-17 01:00 the IMU puts the antenna at −66° elevation (43 files), against −2.2° hanging at 30 m on 07-16. That is the box tilted on the ground for tests, as memo 001 records, not hanging; the elevation returns near 0° at 01:00, when the raise begins.
- **Resting against hanging.** Hanging at 30, 87.5 or 91 m, the elevation moves within every file: 0% of those files have a spread under 0.1°, and the file medians take 99–166 distinct values. Before the 07-15 lift and on the afternoon of 07-15, 25–29% of files are that still, and the median stays at −3.2° (8 values in 1008 files): resting, not hanging. The 07-15 night itself has IMU data in only 5 of its 470 files and no lidar, so it is bracketed by these stretches, not tested.

The labels hold where these streams can test them. There is no horizon profile at 2 m, so the ground stretches cannot be modelled: that profile (`horizon_profiles_v0005` in memo 002's numbering) is the missing input. Until it exists, the 07-15 night, the 07-17 sunset and the ground half of the 07-18 night are left out of every fit here.

window (UTC)	mode table	lidar median (m)	IMU el. (deg)	IMU spread (deg)	files
07-14 00:00-07-15 03:30	∼2m	0.2 (0.2–0.2)	−3.2	0.25	1363
07-15 03:30-14:00, the 07-15 night	∼2m	–	−3.2	2.00	470
07-15 17:00-21:28	∼2m	–	−3.0	0.24	124
07-15 23:00-07-16 16:30	∼30m	–	−2.2	0.48	482
07-16 23:27-07-17 01:00, the 07-17 sunset	∼2m	–	−66.2	3.63	44
07-17 01:00-02:30, the raise	raising	–	0.4	2.91	42
07-17 04:00-16:00	∼87.5m	–	−0.0	0.46	336
07-17 16:00-21:28	∼91m	–	−0.2	0.92	153
07-17 23:30-07-18 00:00	unknown	1.1 (0.7–1.6)	−−	–	14
07-18 00:00-00:30	unknown	30.6 (27.8–39.4)	−−	–	14
07-18 00:30-03:10	∼91m-reraise	250.0 (250.0–250.0)	−176.0	0.71	71

Table 2: Box-air's lidar and IMU per window (`validate_heights.json`): the mode-table label, the lidar's median distance (10th–90th percentile, medians over files), the IMU's median elevation and the median within-file spread, and the number of raw files.

5 The night fit

`fit_v0017` fits Eq. 1 without the Sun, to the 165 bins with the Sun below −10° (Fig. 3, Table 3). All of them are rx-A, at one pointing and one height.

- T_{gnd} **is not measured.** It is exactly degenerate with the offset, as Section 3 predicts. At 25 frequencies the ±1° elevation wobble formally separates them. The values there run from −7053 to 11022 K: fitted model error, not a temperature. The pedestal $\langle f_{\mathrm{gnd}}\rangle T_{\mathrm{gnd}} + T_{\mathrm{off}}$ is determined: 12 to 90 K at 110–170 MHz.

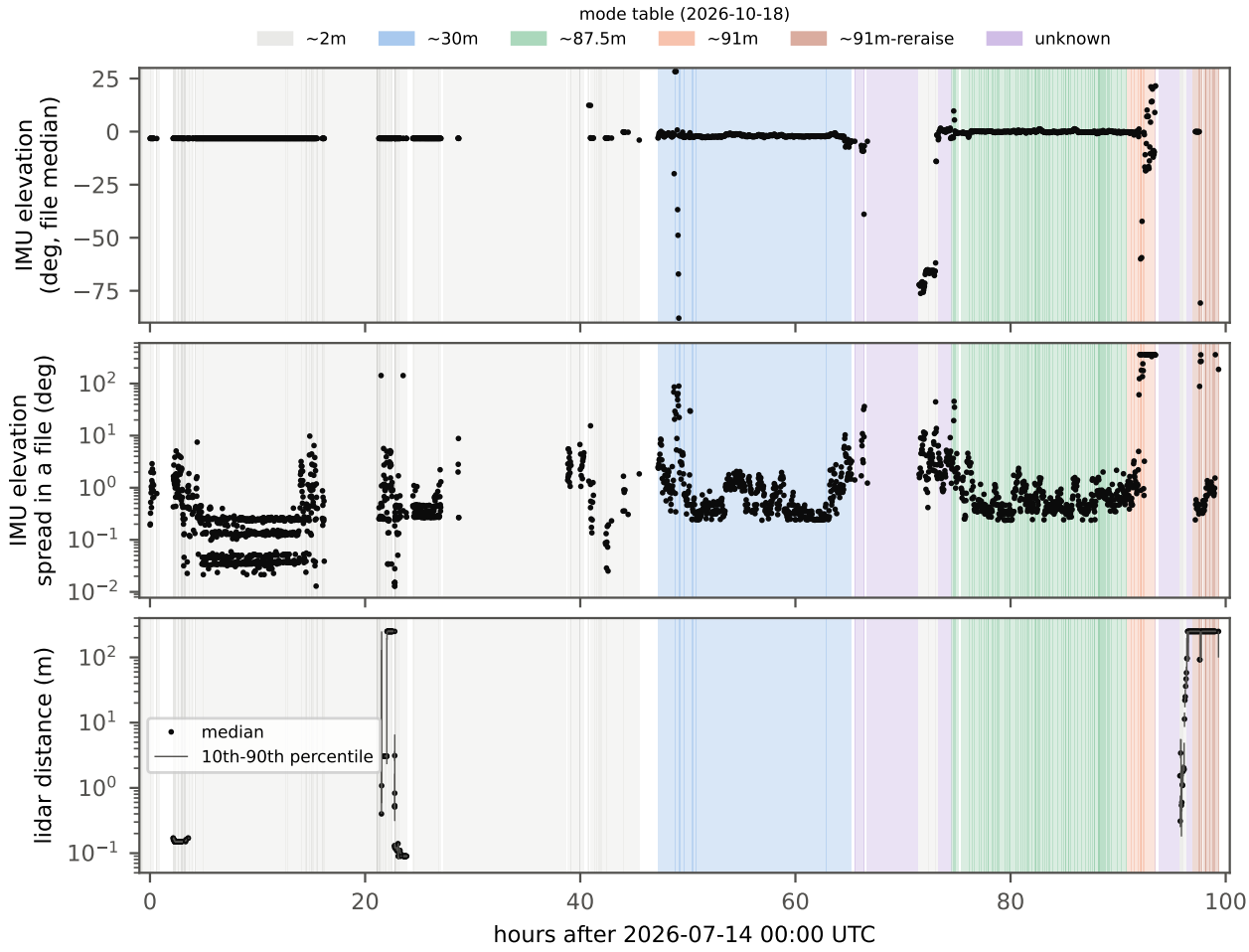


Figure 2: Box-air's height from its own sensors, 07-14 to 07-18, one point per raw file (`validate_heights.npz`), against the 2026-10-18 mode table (shading). Top: the IMU's median elevation in each file (readings near ±180°, IMU wraps, fall off the axis). Middle: its spread within the file (max − min), log scale. Bottom: the lidar's median distance (points) and 10th–90th percentile (lines), log scale, where it returned any. Hanging at height the elevation always moves; on the ground before the lift it often does not, and at the 07-17 sunset it sits near −66°. The lidar sees the 07-18 lowering and re-raise.

- **The sky amplitude is** $a \approx 2.1$, measured to ± 0.02 at each frequency. Two explanations remain open: the `tcal` kelvin scale (it assumes the nameplate noise-source excess, memo 004), or a sky fraction that HFSS underestimates. The fit cannot tell them apart.
- **The model is not adequate at the noise level**: median χ^2 per bin is 120, and 234 on held-out alternating 30-min blocks.
- **The morning behind the ridge looks like night.** Adding the bins from 0° sunrise to the ridge crossing, `fit_v0018` (264 bins), raises the median χ^2 only from 120 to 136.

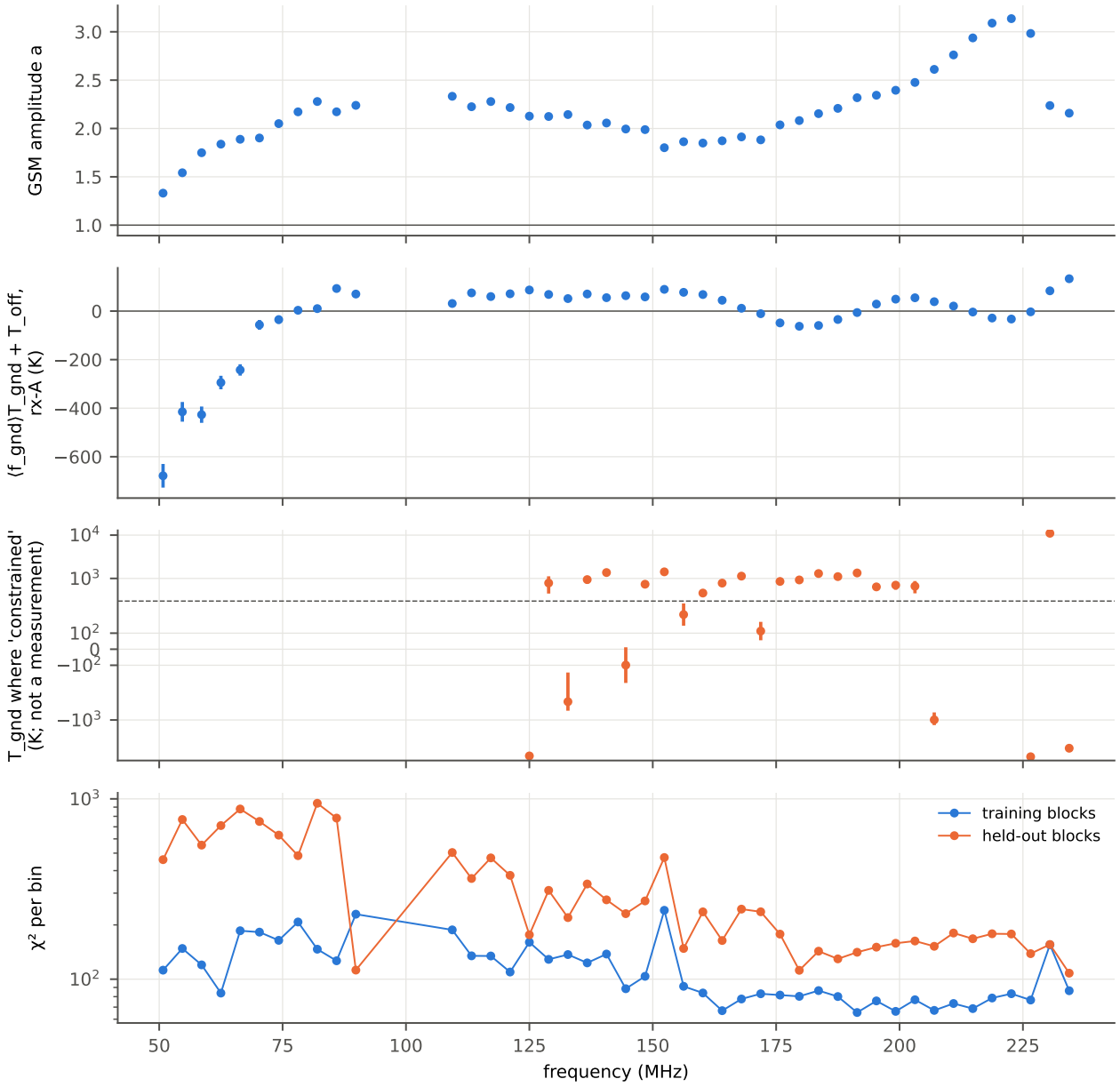


Figure 3: The night fit (`fit_v0017`). From the top: GSM amplitude; the rx-A pedestal; T_{gnd} where the solver calls it constrained (symlog axis; the dashed line is 300 K); χ^2 per bin on training and held-out blocks.

product	bins fitted	bins	freqs	median χ^2	median held-out χ^2
`fit_v0017`	night (Sun below −10°)	165 / 302	44	120	234
`fit_v0018`	Sun behind the ridge allowed	264 / 302	44	136	254
`fit_v0019`	night, 60 s bins	319 / 615	44	110	240
`fit_v0020`	Sun modelled, all bins	302 / 302	44	486	35520

Table 3: The fit products. Every fit evaluates the model on all bins; "bins fitted" are those used to solve. `fit_v0020` fits the Sun column on every bin, including those with the transmitter on; its held-out χ^2 reaches 3×10^6, and it is not used.

6 The Sun

6.1 When it is visible

Table 4 lists every time the Sun's centre crosses the DEM horizon or 0° altitude while each height era was flown in phase C. Figure 4 shows the horizons and the Sun's tracks. At 87.5 m the Sun clears the eastern ridge at 15:54 UTC on 07-17, at 39° altitude and azimuth 94°, with the edge 352 m away. That is 3.5 h after it reaches 0° altitude at 12:26. At 91 m it sets behind the western ridge at 01:21 on 07-18 (17°, edge 171 m), 1.5 h before 0° (02:53).

Two consequences follow. The rx-B window starts at 01:27, after the terrain sunset, so of the calibrated data only 07-17 15:54–16:22 sees the direct Sun. And the 2026-10-01 trials' observation that the Sun "lifts 199 MHz within an hour of rising" cannot be the direct Sun: that hour is behind the ridge.

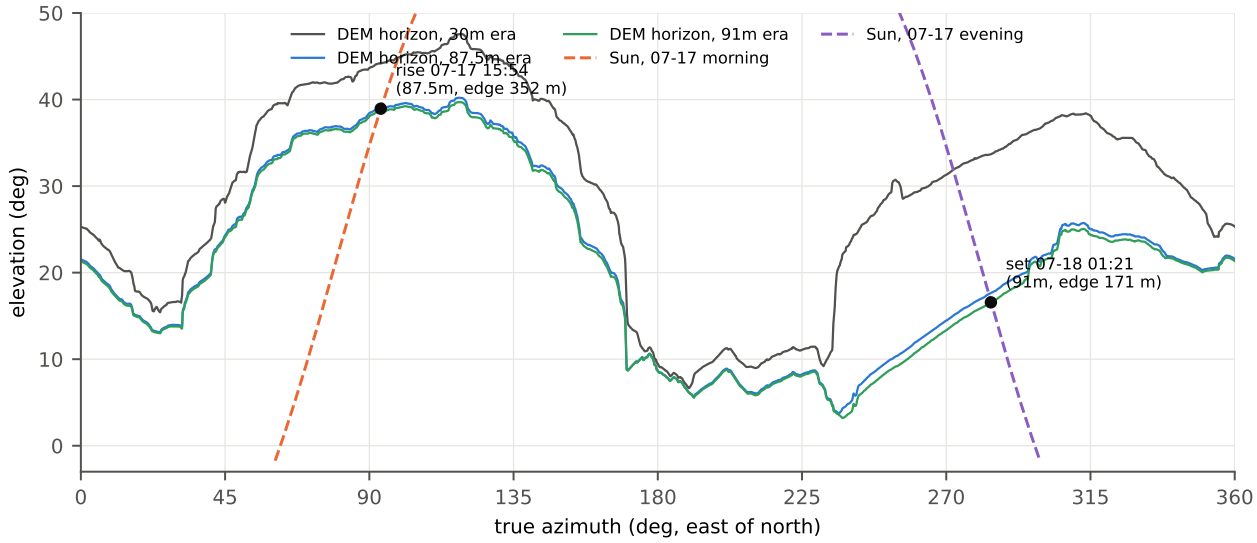


Figure 4: DEM horizon against true azimuth for the three heights (`horizon_profiles_v0004`, rotated from grid to true bearings), and the Sun's tracks on 07-17. Dots mark the terrain crossings in the calibrated eras.

era	event	horizon	UTC	Sun alt (deg)	Sun az (deg)	horizon (deg)	edge (m)
30m	set	terrain	07-16 00:01:47	32.0	272.6	32.0	151
30m	set	flat	07-16 02:54:17	-0.0	298.1	0	–
30m	rise	flat	07-16 12:25:27	0.0	62.0	0	–
30m	rise	terrain	07-16 16:24:17	44.8	98.7	44.8	335
87.5m	set	flat	07-17 02:53:44	-0.0	297.9	0	–
87.5m	rise	flat	07-17 12:26:14	0.0	62.3	0	–
87.5m	rise	terrain	07-17 15:54:34	38.9	93.6	38.9	352
91m	set	terrain	07-18 01:21:27	16.6	283.9	16.6	171
91m	set	flat	07-18 02:53:07	-0.0	297.6	0	–

Table 4: Sunrises and sunsets during phase C, by height era (`sun_events_v0005`). "terrain": the Sun's centre crosses the DEM horizon traced along its bearing; "flat": 0° altitude. Geometric, with no refraction.

6.2 The test around 15:54

The test reads `fit_v0019`, a night fit on one-minute bins. Every bin with the Sun above 0° is out of sample there, so its residual (data minus fitted sky, ground and offset) is a prediction. That residual drifts by tens of kelvin over hours, as the night model is carried into the morning. So each frequency is fitted locally, from 14:30 to 16:23 UTC, as

$$\mathrm{resid}(t) = c_0 + c_1(t - t_{\mathrm{ridge}}) + S_\odot\, A_\odot(t - \tau).$$

The transmitter of the beam-mapping raster is on 15:36–16:14, with short gaps (box-gnd transitions, memo 001). Its bins are never fitted. That leaves 62 bins, of which only 7 (16:14–16:22) have the Sun above the ridge. Errors on $S_\odot$ are scaled by $\sqrt{\chi^2/\mathrm{dof}}$.

Each of the following checks could have come out the other way:

1. **A null at 0° sunrise.** The same fit around 12:26 (97 transmitter-off bins), with a Sun column that ignores the terrain, should give $S_\odot \approx 0$ if the ridge hides the Sun.
2. **Timing.** χ^2 against a shift τ of the Sun column should be smallest near $\tau = 0$.
3. **Spectrum.** $S_\odot(\nu)$ should be positive and smooth, of the order of the quiet-to-active Sun.

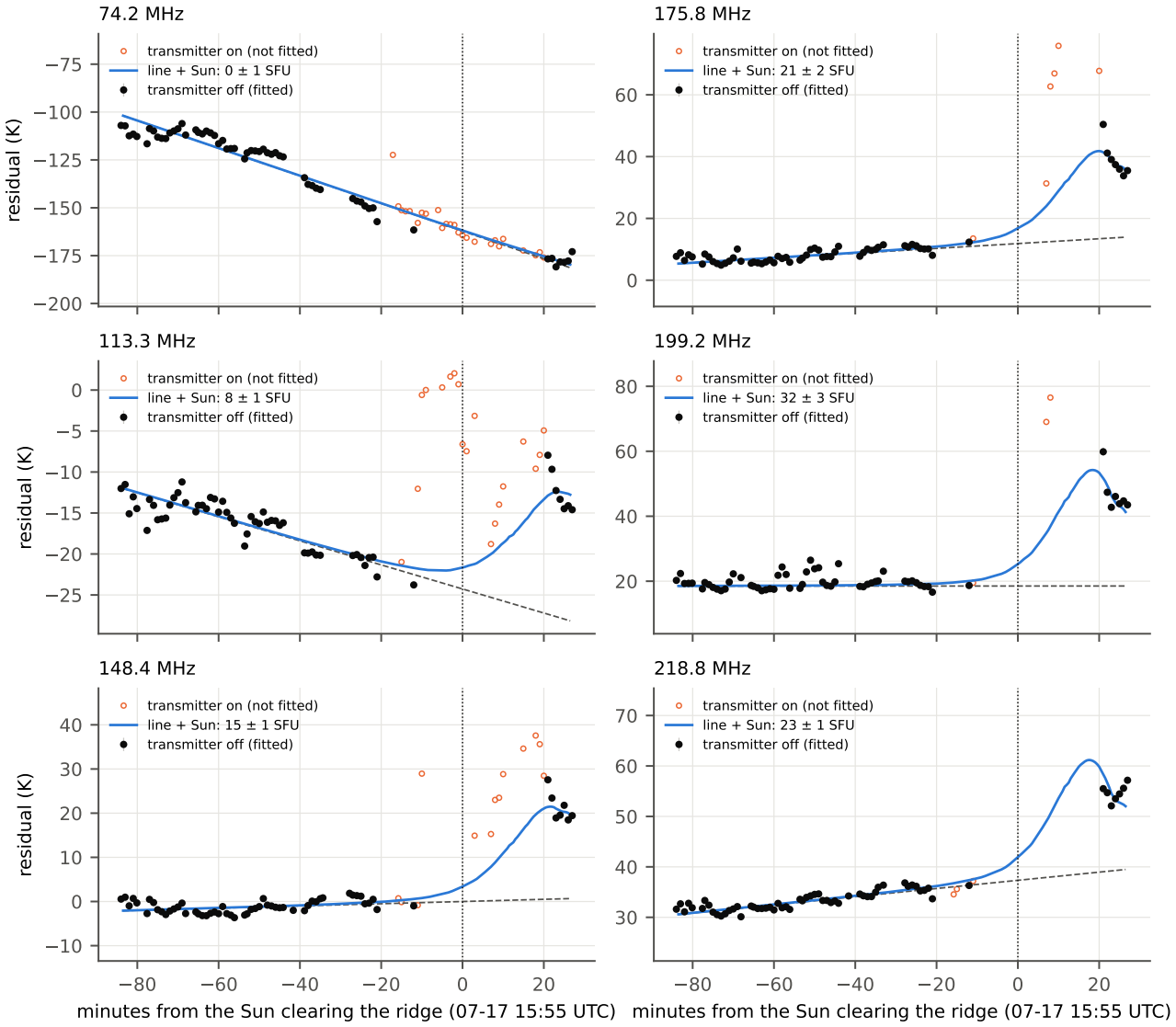


Figure 5: Out-of-sample residuals of the night fit around the ridge crossing (dotted line), at six frequencies. Filled: transmitter off, fitted. Open: transmitter on, not fitted. Dashed: the fitted line alone. Blue: line plus Sun.

6.3 Result

- **Spectrum.** At 31 frequencies between 105 and 235 MHz, $S_\odot$ runs from 7 to 33 SFU (31 positive), rising smoothly: 7±1 at 109 MHz, 15±1 at 148, 32±3 at 199 and 23±1 at 219 MHz (Fig. 7, Table 5).
- **Null.** At 0° sunrise the same fit gives −6 to 1 SFU over the same band. No step appears when the Sun rises behind the ridge.
- **Timing.** χ^2 is smallest at $\tau = -3$ min, and within 5 of that minimum from −7 to +3 min. A shift of 20 min earlier costs $\Delta\chi^2/(\chi^2/\mathrm{dof}) = 69$ on average.
- **Kelvin scale.** If the factor $a \approx 2$ lies in the kelvin scale or the beam, it scales $S_\odot$ too. $S_\odot/a$ is 3, 8 and 14 SFU at 109, 148 and 199 MHz.
- **Below 90 MHz** the result is not usable. The residual drifts by tens of kelvin across the window there, and the null fails (grey and orange points below 90 MHz in Fig. 7).

The spectra themselves. Figure 6 sets aside the local fit and compares spectra. It averages the calibrated spectrum (weighted means) over two windows of transmitter-off bins. "Before" is −40 to −15 min from the ridge crossing, with the Sun behind the ridge (15 bins). "After" is the 7 bins with the Sun above it. Their difference is shown raw, and with the night model's predicted sky and ground change between the windows subtracted. That change is at most 9 K above 109 MHz, but up to 51 K below 87 MHz, where the difference is dominated by drift the model does not capture. The difference is compared with a Sun predicted with nothing fitted to these data. The prediction takes RSTN's concurrent 245 MHz flux (§6.4), extrapolated down in frequency with the MWA quiet-Sun shape (scaled by 0.85), times the Sun column averaged over each window.

- Above 109 MHz the spectrum rises by a smooth 20 K at 148 MHz and 25±1 K at 199 MHz. The flux this implies, 15 and 34 SFU, agrees with the local fit.
- The prediction is 5 and 5 K. The measured jump is 1.9–5.6 times the prediction over 31 frequencies (weighted mean 3.6). Scaled by a, the prediction falls short by a factor of 2.1 below 200 MHz and 1.2 above 205 MHz. The excess is flatter in frequency than the quiet-Sun shape.
- **Not a gain change.** A change in receiver gain would scale the whole spectrum, so $\Delta T/T$ would be constant. Instead it runs from 1 to 16% (coefficient of variation 0.49), while ΔT itself varies less (0.27). The jump is additive, like a new source, not multiplicative.

MHz	$S_\odot$ (SFU)	χ^2/dof	a	$S_\odot/a$	null at 12:26
109.4	6.9 ± 0.7	119	2.25	3.1	-1.2 ± 0.5
125.0	13.3 ± 1.2	207	2.11	6.3	-1.0 ± 0.6
148.4	14.5 ± 0.9	125	1.91	7.6	-2.9 ± 1.1
175.8	20.9 ± 1.6	79	1.98	10.5	-2.9 ± 1.7
199.2	32.4 ± 2.5	128	2.36	13.7	-2.9 ± 1.4
218.8	23.5 ± 1.4	125	2.96	7.9	-1.6 ± 1.2
234.4	25.6 ± 1.8	39	2.20	11.7	1.1 ± 1.2

Table 5: $S_\odot$ at selected frequencies: the local fit's χ^2/dof, the GSM amplitude a of `fit_v0019`, $S_\odot/a$, and the flat-horizon null at 12:26.

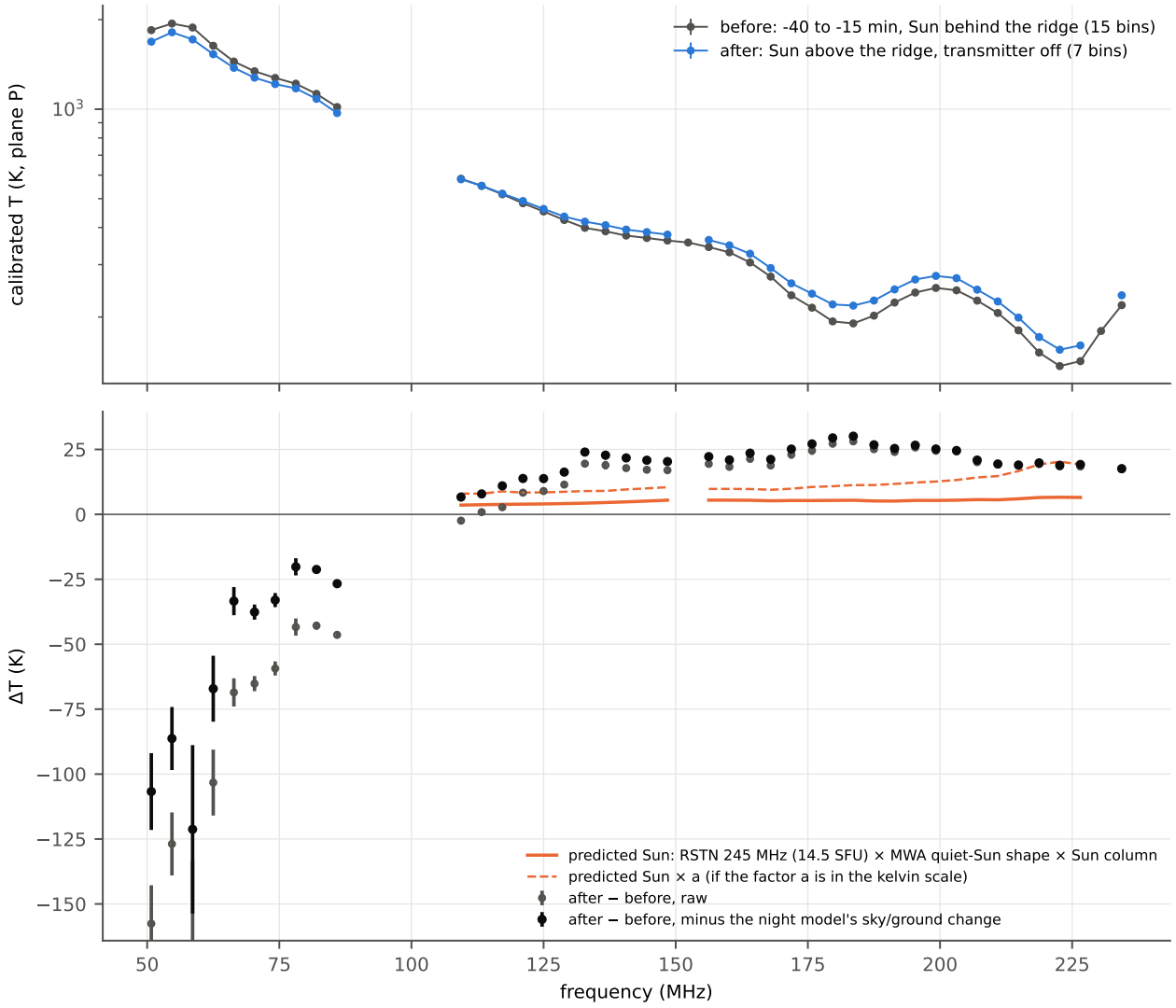


Figure 6: Top: calibrated spectra before the ridge crossing (Sun behind the ridge) and after it (Sun above, transmitter off). FM is blank. Bottom: their difference, raw (open) and with the night model's sky/ground change removed (filled). Orange: the Sun predicted from concurrent RSTN 245 MHz flux with the MWA quiet-Sun spectral shape, through the beam and the ridge; dashed: the same times the GSM amplitude a. Nothing in the orange curves is fitted to these data.

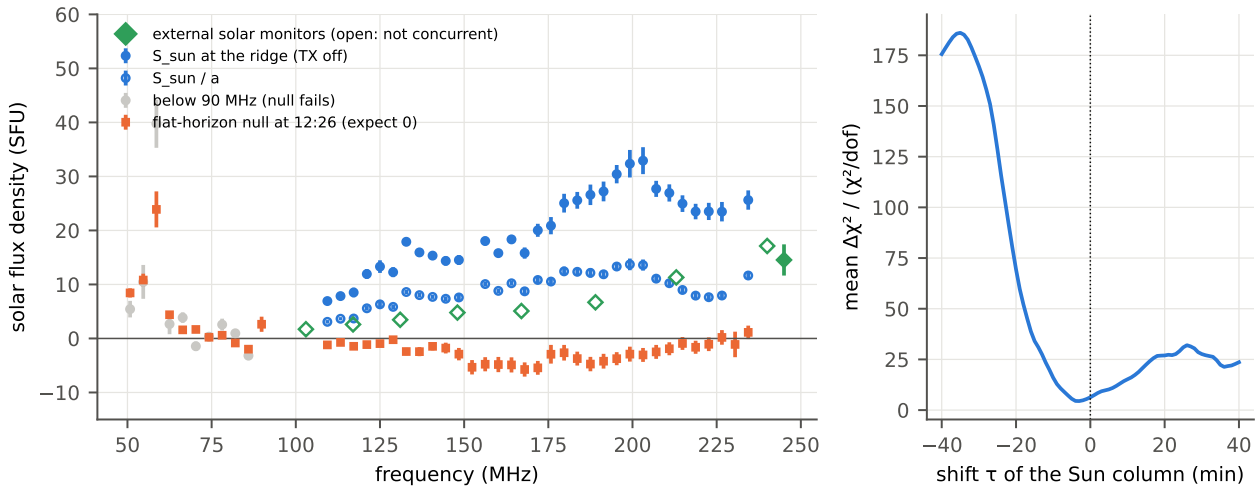


Figure 7: Left: $S_\odot$ from the transmitter-off fit at the ridge crossing (filled), divided by the GSM amplitude a (open), and the flat-horizon null at 12:26 (squares). Diamonds are external solar monitors (Section 6.4). Right: mean $\Delta\chi^2$ over frequencies against a shift of the Sun column; $\tau > 0$ is later than the DEM predicts.

6.4 Concurrent solar monitoring

The USAF Radio Solar Telescope Network (RSTN) measures the whole-Sun flux at 245, 410 and 610 MHz every second. Two stations recorded continuously through 15:54–16:22 UTC on 07-17. At 245 MHz Sagamore Hill reads 14.9 SFU (rms 0.8, 99th percentile 18) and San Vito 14.1 SFU (rms 0.7, 99th percentile 17), over 1651 and 1650 one-second samples. Palehua's reading is below 5 SFU for 52% of the window, as if not yet tracking the Sun; over the transmitter-off bins with the Sun up it reads 25 SFU. The stations' noon values for 07-17 at 245 MHz are Sagamore Hill 13, San Vito 13, Palehua 21, Learmonth 23 SFU. The stations disagree by more than the quoted 2–3 SFU; we take the two continuous stations' mean, 14.5 SFU, as the concurrent reference at 245 MHz, and note that Palehua reads higher.

The Sun was quiet: SWPC lists no flare or radio burst between 13:18 and 20:16 UTC, the 10.7 cm flux was 104 sfu (Penticton, 17:00 UTC), and e-CALLISTO spectrographs show no burst at 100–250 MHz in the window. Sagamore Hill's spectrograph recorded a series of type III bursts at 25–180 MHz from 12:14 to 13:18, inside the null window, when the Sun was behind the ridge. We have not looked for them in our data.

Nothing calibrated was found below 245 MHz for that day. For the spectrum's shape, the MWA's quiet-Sun measurement (Oberoi, Sharma & Rogers 2017, on 2013-09-03) gives 1.7, 3.5, 5.1, 11.3 SFU at 103, 131, 167, 213 MHz, rising with frequency as ours does, with 10–60% systematic error. At 234 MHz, the top of our band, $S_\odot$ is 26±2 SFU and $S_\odot/a$ is 12 SFU. RSTN gives 14.5 SFU at 245 MHz, a ratio of 1.8 ($S_\odot$) or 0.8 ($S_\odot/a$). Between 105 and 200 MHz, $S_\odot$ is 3.0–5.0 times the MWA quiet-Sun curve, and $S_\odot/a$ is 1.4–2.4 times it. The excess thus agrees with the concurrent Sun to within the unresolved factor $a \approx 2$. That agreement is not a test that separates the Sun from another effect of similar size at sunrise.

source	when	what it says
RSTN Sagamore Hill, 245 MHz	15:55–16:22	14.9 SFU mean, rms 0.8
RSTN San Vito, 245 MHz	15:55–16:22	14.1 SFU mean, rms 0.7
RSTN Palehua, 245 MHz	16:14–16:22	25.0 SFU mean; below 5 SFU (as if not tracking) for 52% of 15:55–16:22
RSTN noon values, 245 MHz	07-17 local noon	Sagamore Hill 13, San Vito 13, Palehua 21, Learmonth 23 SFU
Penticton F10.7	17:00	103.9 sfu observed
SWPC event list	all day	0007-0008 PAL RBR 245 820; 0150-0150 PAL RBR 245 260; 1214-1318 SAG RSP 025-180 VI/2; 2016-2053 SAG RSP 025-076 VI/2; 2025-2114 G19 XRA 1-8A B6.5 1.8E-03; 2030-2034 PAL RSP 025-075 III/2; 2214-2251 PAL RSP 025-055 VI/2; 2325-2325 PAL RSP 025-037 III/2
e-CALLISTO (6 stations)	15:45–16:30	no burst at 100–250 MHz (uncalibrated, visual)
MWA quiet Sun (Oberoi et al. 2017)	2013-09-03	103 MHz 1.71, 117 MHz 2.6, 131 MHz 3.46, 148 MHz 4.79, 167 MHz 5.08, 189 MHz 6.69, 213 MHz 11.3, 240 MHz 17.12 SFU

Table 6: External solar radio measurements for 2026-07-17 (`inputs/solar_reference.json`, with URLs).

6.5 Solar radio bursts in the raw data

A radio burst is a property of the Sun, recorded independently by other observatories. It is the most direct test of whether box-air receives solar emission. Type III bursts drift from high to low frequency within seconds and last tens of seconds; at 0.54 s per integration, box-air resolves them.

The reference list. `curation/solar_bursts_v0001.json` lists the 32 bursts that e-CALLISTO spectrographs recorded at two or more stations while the Sun was above the Marjum ridge (Table 4). It starts from e-CALLISTO's curated burst list for July 2026, a detector that requires three stations, with times to the minute. Spectrograms from 11 stations in daylight were then inspected to time each burst to about a second and to add bursts the list missed. All are type III or groups of type III. The stations cover 5–93 MHz, apart from one noisy one at 46–226 MHz, so the list cannot say whether a burst reached our higher frequencies.

Method. For each burst, `sun_bursts.py` reads raw box-air antenna rows from 90 s before to 90 s after the burst, with *no* RFI flags, since the flags mark positive excesses, which is what a burst is. It divides each channel by its median over that cutout. It then measures the excess in four bands during the burst, in robust standard deviations of the same band away from it, and finds when each 10 MHz sub-band peaks. Box-air antenna data exist at 12 of the 32 bursts; the rest fall in data gaps or calibration cycles (Table 7).

What the raw data show. Figure 8 shows three bursts as raw dynamic spectra, each channel divided by its median.

- **07-15 18:33 (on the ground).** A narrow ridge of emission appears inside the e-CALLISTO interval and drifts from the 70 MHz range down to the bottom of our band. The 10 MHz sub-bands peak in sequence, at −8 MHz/s. In the light curves, 45–65 MHz rises by 99% of the raw power (115σ), while 170–235 MHz does not move (1σ).
- **07-15 18:23 (on the ground).** Two components about 25 s apart, both low-frequency only, 119σ at 45–65 MHz. The brighter, later one drifts at −9 MHz/s.
- **07-17 20:33 (91 m, during the raster).** The brightest burst of the period (e-CALLISTO's "IIIG/3", a group). Several components sweep down to the bottom of our band. Sub-bands up to 120 MHz peak above 8σ and up to 220 MHz above 5σ, so the group reaches above the reference stations' bands. The raw power at 45–65 MHz rises by 120%. The raster's own modulation is visible behind it.

Test. A random time in daytime is often near one of the impulsive spikes of §6.6, so a big excess alone is not a detection. The bursts e-CALLISTO saw are bright at 45–85 MHz and fainter above, while the impulsive

spikes are broadband. The test therefore counts times that are bright at 45–65 MHz *and* quiet at 170–235 MHz ($z_{45–65} > 10$, $z_{170–235} < 5$), at the bursts and at 250 random times in the same Sun-up windows, at least 5 min from any listed burst (Fig. 9). 5.2% of the random times pass, against 5 of the 12 bursts with data; the chance of that many by accident is 2.2×10^{-4}. This test could have failed.

What this shows. Box-air records solar radio bursts. They come at the times independent stations recorded them, drift from high to low frequency as type III bursts do, and have the low-frequency-weighted spectrum those stations saw. That is a physical identification of solar emission in our data, not a coincidence in timing. It says nothing about the quiet Sun's flux, since the e-CALLISTO spectrograms are not calibrated.

burst start (UTC)	dur. (s)	e-C MHz	sta.	box-air excess (robust σ) 45–65	65–85	110–170	170–235	match
07-15 18:21:37	14	20–85	4	6	3	16	1	no
07-15 18:23:19	43	20–85	5	119	36	10	2	yes
07-15 18:33:02	19	10–85	6	115	4	1	1	yes
07-15 18:47:01	12	10–45	2	no box-air antenna data				–
07-15 22:45:10	5	10–45	2	no box-air antenna data				–
07-15 22:59:08	8	10–40	2	no box-air antenna data				–
07-15 23:49:00	10	15–50	2	11513	1796	3046	2184	no
07-16 16:23:55	19	20–85	6	19	1	2	4	yes
07-16 16:50:02	11	20–85	6	no box-air antenna data				–
07-16 16:54:58	13	20–90	7	61	11	72	19	no
07-16 17:16:06	26	20–85	6	no box-air antenna data				–
07-16 17:25:35	5	46–85	3	no box-air antenna data				–
07-16 17:26:22	10	20–85	6	no box-air antenna data				–
07-16 17:57:21	50	46–80	3	no box-air antenna data				–
07-16 18:33:13	31	20–85	6	no box-air antenna data				–
07-16 18:48:05	11	20–65	3	no box-air antenna data				–
07-16 19:17:07	16	20–85	5	no box-air antenna data				–
07-16 21:36:13	8	20–55	2	no box-air antenna data				–
07-16 22:05:44	10	15–45	2	no box-air antenna data				–
07-16 22:25:10	9	15–50	2	no box-air antenna data				–
07-16 22:28:10	16	15–60	3	no box-air antenna data				–
07-16 22:32:42	9	55–85	3	no box-air antenna data				–
07-16 23:58:59	34	20–75	2	17	9	16	20	no
07-17 20:16:26	14	30–85	4	44	9	3	0	yes
07-17 20:30:45	46	10–60	5	8	2	2	4	no
07-17 20:32:59	37	10–90	6	35	10	5	4	yes
07-17 20:34:14	23	10–85	5	5	2	0	1	no
07-17 21:21:22	99	10–60	4	4	4	2	2	no
07-17 22:26:42	49	15–45	2	no box-air antenna data				–
07-17 22:46:54	19	20–60	4	no box-air antenna data				–
07-17 22:51:02	21	15–40	2	no box-air antenna data				–
07-17 23:25:37	17	15–45	2	no box-air antenna data				–

Table 7: The 32 reference bursts (`curation/solar_bursts_v0001.json`): duration and frequency range at the e-CALLISTO stations, number of stations, and box-air's excess in four bands. "match": bright at 45–65 MHz and quiet at 170–235 MHz ($z_{45–65} > 10$, $z_{170–235} < 5$).

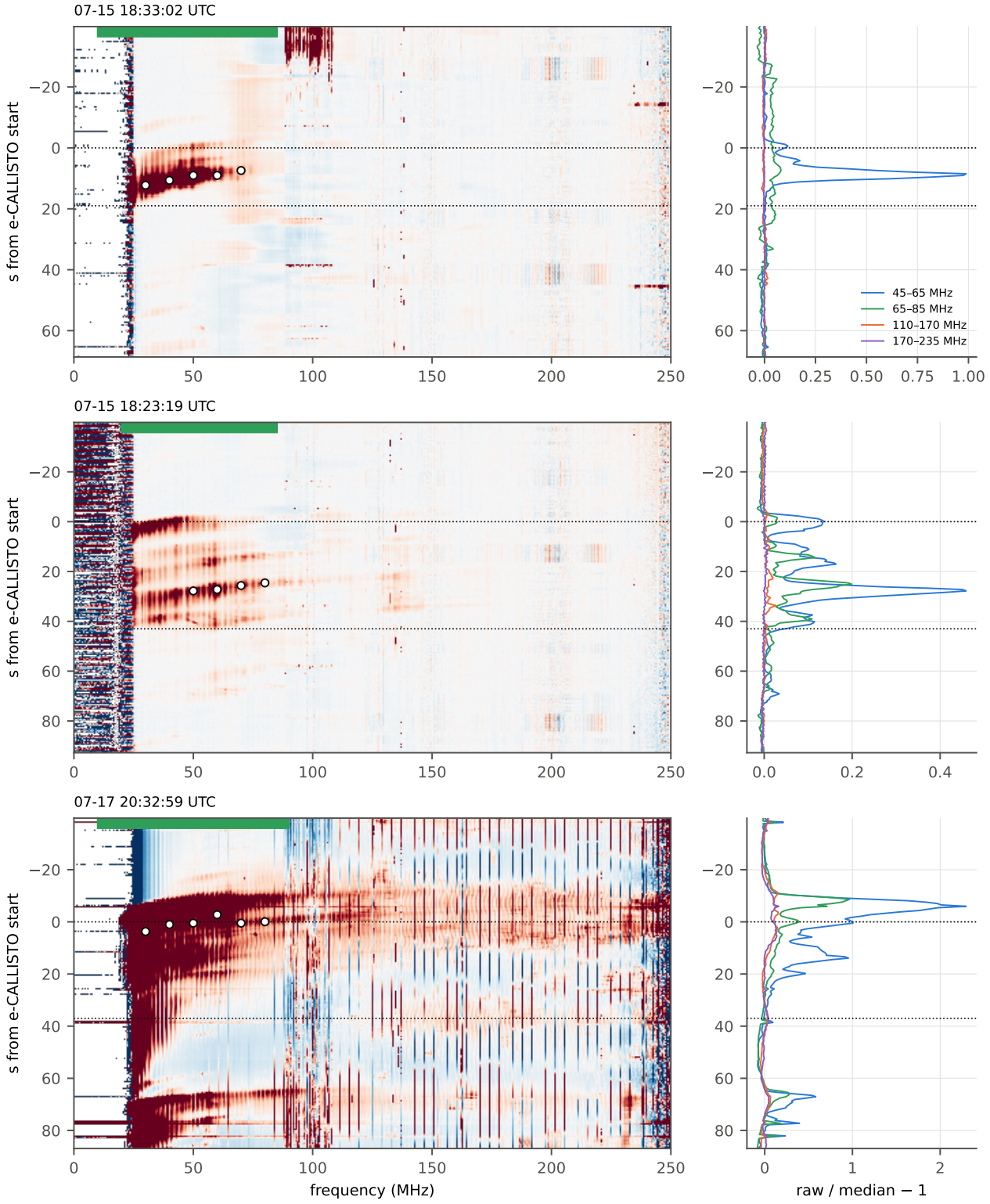


Figure 8: Three e-CALLISTO bursts in raw box-air data, time running down. Left: each channel divided by its median over the cutout, minus 1. Dotted lines are the burst's start and end at the reference stations, and the green bar along the top is their frequency range. Circles mark when each 10 MHz sub-band below 95 MHz peaks, where it is clear (above 8 robust σ) and within 8 s of the 45–65 MHz peak. Right: the same in four bands, on the same time axis.

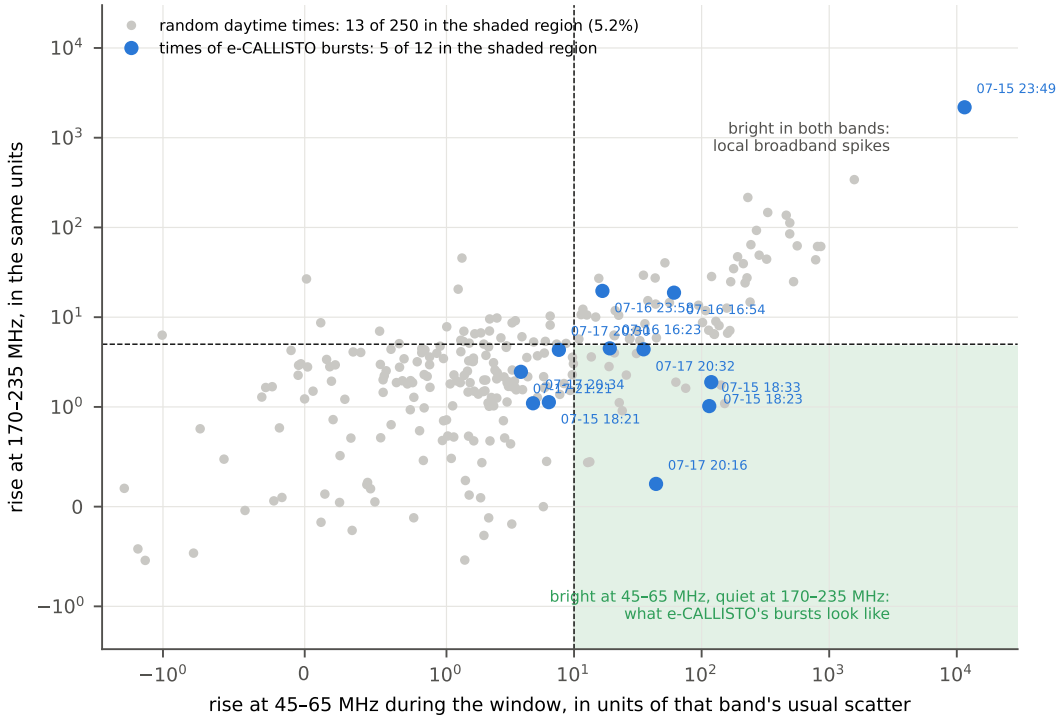


Figure 9: The burst test. Each point is one 15–40 s window of raw box-air data. Its position says how much the raw power rose during the window, at 45–65 MHz (across) and at 170–235 MHz (up), in units of that band's usual scatter. Blue: the windows of the e-CALLISTO bursts, labelled by start time. Grey: 250 random windows in the same daytime hours. The bursts e-CALLISTO recorded are bright at low frequency and faint above 170 MHz, so a real one should land in the shaded region ($z_{45–65} > 10$, $z_{170–235} < 5$). The impulsive spikes of §6.6 rise in both bands and land at the upper right. 5 of 12 bursts fall in the shaded region, against 5.2% of the random windows. Axes are linear between −1 and 1 and logarithmic beyond.

6.6 Impulsive events in the raw data

Looked at without a reference list, raw box-air is full of impulsive events: single integrations that rise at every frequency at once, about 103 times an hour in daytime (Fig. 10, as horizontal lines). They do not drift. `impulsive_scan.py` finds them in phase C (35 h of antenna data at confirmed heights) and sorts them by whether the Sun is above the ridge at the antenna, up but behind the ridge, or down. Overall they come 103 per hour with the Sun visible (7.2 h), 75 with it behind the ridge (11.2 h), and 7 at night. Earlier versions found the visible and behind-the-ridge rates equal and called the events local; that comparison included the stretches on the ground, where they were most frequent.

These classes differ in height era and time of day as well as in the Sun, so `validate_impulsive.py` holds one variable at a time, on exposure rebuilt row by row in 1-minute cells (Table 8). Events cluster, so beside exact Poisson intervals it gives a bootstrap over 30-minute chunks, which is far wider and the one to use.

- **Within one era** the comparison goes both ways. At 30 m the rate is 169 per hour with the Sun visible against 69 behind the ridge, but the latter is 153 on 07-15 and 12 on 07-16, so one day carries it. For events of three or more integrations the visible rate is higher on both days (21 and 32 against 9 and 6). At 91 m, one evening, it is the other way: 144 visible against 272 behind the ridge (25 against 42 for the longer events).
- **The same hours of local time** with the Sun visible and behind the ridge overlap in only 2 hours (10:00, 19:00 MDT). Pooled over them, the visible rate is 3.5 times the other (1.7 for the longer events), but at 19:00 at 91 m alone the ratio is 1.65 (0.57 for the longer events).
- **Against the Sun's height above the horizon**, at 30 m the rate shows no step where the Sun crosses it: 198 per hour with the Sun 5–10° behind the terrain, 39 within 2° below it and 101 within 2° above. At 91 m it peaks within 2° of the horizon (608 and 795 per hour) and is 88 with the Sun high, but that is one sunset evening.

The controls disagree with each other and between eras, and several rest on one day or two chunks. They neither show the events following the Sun nor rule it out, so their origin is open. That they are broadband single integrations that do not drift, unlike the bursts of §6.5, is why a search for bursts must start from an independent list.

era	Sun	hours	events	rate, all (h^{-1})	rate, 3+ integrations (h^{-1})
30 m	visible	1.7	279	169 (149–190; 104–225)	27 (19–36; 16–35)
30 m	behind ridge	6.6	451	69 (62–75; 16–135)	6.8 (5.0–9.2; 2.8–11.2)
30 m	night	9.5	68	7.1 (5.5–9.1; 1.9–15.4)	0.4 (0.1–1.1; 0.0–1.1)
87.5 m	visible	2.8	38	14 (10–19; 1–38)	4.7 (2.5–8.0; 0.0–13.6)
87.5 m	behind ridge	3.1	24	7.7 (4.9–11.5; 1.9–15.1)	1.3 (0.3–3.3; 0.0–3.4)
87.5 m	night	7.5	40	5.4 (3.8–7.3; 1.8–10.3)	0.5 (0.1–1.4; 0.0–1.7)
91 m	visible	2.9	414	144 (130–158; 60–274)	25 (20–32; 8–51)
91 m	behind ridge	1.3	363	272 (245–302; 185–424)	42 (32–55; 25–82)
91 m	night	0.1	6	48 (18–104; 0–103)	0.0 (0.0–29.3; 0.0–0.0)

Table 8: Impulsive-event rates within each height era by the Sun's position (`validate_impulsive.json`; `impulsive_v0003`, rows at confirmed heights): rate per hour with its 95% exact Poisson interval and, after the semicolon, the 95% bootstrap interval over 30-minute chunks, which allows for clustering. "visible": the Sun above the traced horizon; "behind ridge": above 0° altitude but below it.

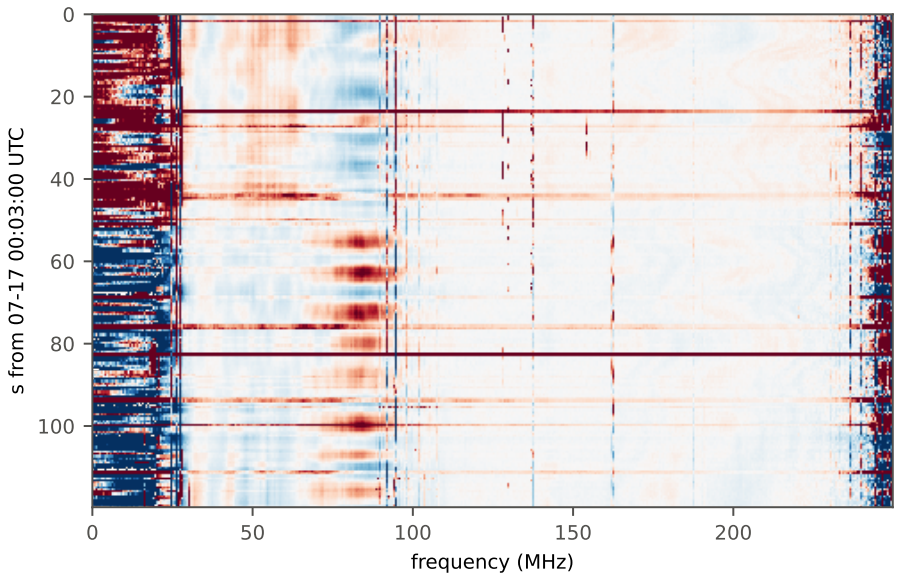


Figure 10: Two minutes of raw box-air data in daytime with the Sun behind the ridge, each channel divided by its median, time running down. The horizontal lines across all frequencies are the impulsive events.

6.7 Sunrise and sunset in the raw data

The calibrated data hold one terrain crossing. Phase C holds 4, the rest uncalibrated (Table 4). `sun_transitions.py` keeps the raw context of each, minute by minute, ±60 min (Figs. 12 and 13): the mean raw antenna spectrum, whether the transmitter was on, whether the antenna moved, and how many rows were antenna rows rather than calibration states.

It then fits raw power at each beam frequency as a line plus the Sun column at that crossing's pointing, over the parked stretch of data containing the crossing, without transmitter-on minutes unless the whole stretch has it on. Raw power has no kelvin scale. The fit gives $S_\odot/T_{\rm sys}$, and the 5th–95th percentile of the calibrated temperatures of 07-17 stands in for $T_{\rm sys}$. That stand-in comes from other hours and heights, so the fluxes in Table 9 are good to a factor of about 2.

The crossings in context. At these pointings the Sun's predicted signature in raw power is not a step but a slow bump. As the Sun sinks towards the ridge, the beam's gain toward it changes, and the ridge then cuts it off over 10–30 min. So it peaks some 20–30 min before sunset (the sunrise version is the blue curve of Fig. 5). It amounts to a few percent of the power for a Sun of a few SFU. Figures 12 (sunsets) and 13 (sunrises) show what the raw data do. Earlier versions also listed the 07-15 sunrise and the 07-17 sunset at 30 m; the antenna was on the ground for both (§4.1), there is no horizon to trace from 2 m, and they are left out.

- **07-16 sunset, 30 m, transmitter off**: the quietest crossing. Raw power rises steadily, by about 10% over the two hours, at all four frequencies. The fit puts a Sun of 8–11 SFU at 199 MHz on top of that rise, mostly from a bump 15–30 min before sunset. The bump is no larger than the unexplained wander elsewhere in the window.
- **07-16 sunrise, 30 m**: box-air's self-EMI comb (memo 006) is on until 16:51, and its lines cover much of the raw spectrum; the fitted fluxes are erratic.
- **07-17 sunrise, 87.5 m**: the transmitter is on for most of 15:36–16:14 (orange ticks), and its leakage dominates those minutes (the 10–30% excursions in the bottom panel). Calibration cycles also interrupt the antenna rows (grey). On transmitter-off minutes, including ones after the calibrated window ends at 16:22, the raw power is flat to about 1%, and the Sun term is 11–16 SFU at 199 MHz.
- **07-18 sunset, 91 m, transmitter on throughout**: fitted on channels midway between the comb teeth. The raw power wanders by several percent, and the Sun term is 7–10 SFU at 199 MHz.

A solar burst at the 07-17 sunset, on the ground. The antenna was on the ground, tilted to about −66°, from 07-16 23:27 to 07-17 01:15 (§4.1), not hanging at 30 m as earlier versions assumed. `validate_sunset_burst.py` reads the raw spectra of both boxes from 23:44 to 00:08 UTC and divides them by their median over the quiet minutes on either side (Fig. 11). From 23:53:45, box-air's power below 100 MHz rises far above that. Over 23:55:30–00:00:00 it is 1.09 times the reference at 50–60 MHz, 1.54 at 66–72 and 2.30 at 78–84 MHz, and 1.18 at 100–120 and 0.99 at 180–230 MHz.

- **It is not interference.** It is smooth across the band, and channels at whole megahertz rise like their neighbours (1.58 against 1.56), so it is not box-air's 1 MHz self-EMI comb (memo 006). Box-gnd, on the same correlator, stays at 0.95–1.05 times its reference below 90 MHz. The excess's lower edge drifts from about 80 to 50 MHz over ten minutes, with a fainter band near 160–175 MHz.
- **It is not pointing.** At 72–84 MHz it peaks at 2.5 times the reference at 23:58, at an elevation of −73°. At 23:44, at the same elevation, the power was 0.97 times the reference.
- **When it ends.** In 10 s means, the excess falls below 1.5 times the reference at 00:00:45 and below 1 at 00:02:35, while the elevation is still wandering about its park value of −73°. The elevation then steps at 00:02:45 and is within 1° of its final −66° from 00:03:25. With the step the power falls by a further 18%.
- **A flare was in progress.** GOES-18 recorded a C1.5 X-ray flare from 23:51 to 00:09 UTC, peaking at 00:02 (SWPC event 7990, `inputs/solar/20260716events.txt`). The same event has a 245 MHz burst of 820 SFU at 00:07–00:08 (Palehua); box-air at 180–230 MHz stays within 4% of its reference from a minute before it to two minutes after. e-CALLISTO stations (HAARP, COHOE) recorded a type III burst at 23:58:59 (34 s, 20–75 MHz) inside the excess (Table 7).

So the excess is solar emission during that flare: it is box-air's alone, broadband and smooth, and it coincides with the flare and with the type III burst. Its slow downward drift and the band near twice its frequency resemble a type II burst, but no observatory's sweep is listed to confirm the type. Why it ends is open. Earlier versions timed its end against the Sun's disk going behind the ridge as seen from 30 m and read it as occultation; from the ground the horizon is higher and not traced (there is no 2 m profile), so that timing is withdrawn. The flare's own decay, the Sun setting behind the terrain as seen from where the box stood, or the tilted antenna's response could each end it. Most of the fall still comes before the elevation moves; only the last 18% comes with the elevation step. That box-gnd misses it is not explained either: its position is not solved, and here both boxes were on the ground.

What the crossings say. Over the 3 fitted crossings without the self-EMI and 121–219 MHz (where above 2σ), the Sun term runs from 3 to 16 SFU. The concurrent prediction (RSTN at 245 MHz with the MWA quiet-Sun shape, Fig. 6) is 2–11 SFU over the same frequencies. RSTN's noon flux at 245 MHz stayed between 13 and 17 SFU from 07-15 to 07-18. So the raw data allow a Sun of the expected size. They do not detect it convincingly: the Sun term is a few-percent bump on raw power that drifts by as much for other reasons, and the kelvin stand-in is good only to a factor of 2. What they do show is that no crossing has a jump as large as the calibrated 07-17 excess implies, which is 2.0–3.2 times larger than their upper bounds (32 SFU at 199 MHz). That includes the same 07-17 crossing in raw data: with a longer stretch of transmitter-off minutes, it gives 11–16 SFU at 199 MHz.

The timing is weakly constrained. The best shifts τ^* in Table 9 range from −60 to −4 min, with broad χ^2 minima; 1 of them sits at the ±60 min edge of the scan, so its shift is not bounded. The ground under the antenna goes into shadow about half an hour before the antenna at the 30 m sunset and nearly two hours before at the 91 m one, and comes into sunlight 15–50 min after it at the sunrises ("ground" column). A signal from the lit ground, or from heating there, would therefore lead at sunset and lag at sunrise. These data cannot tell those timings apart.

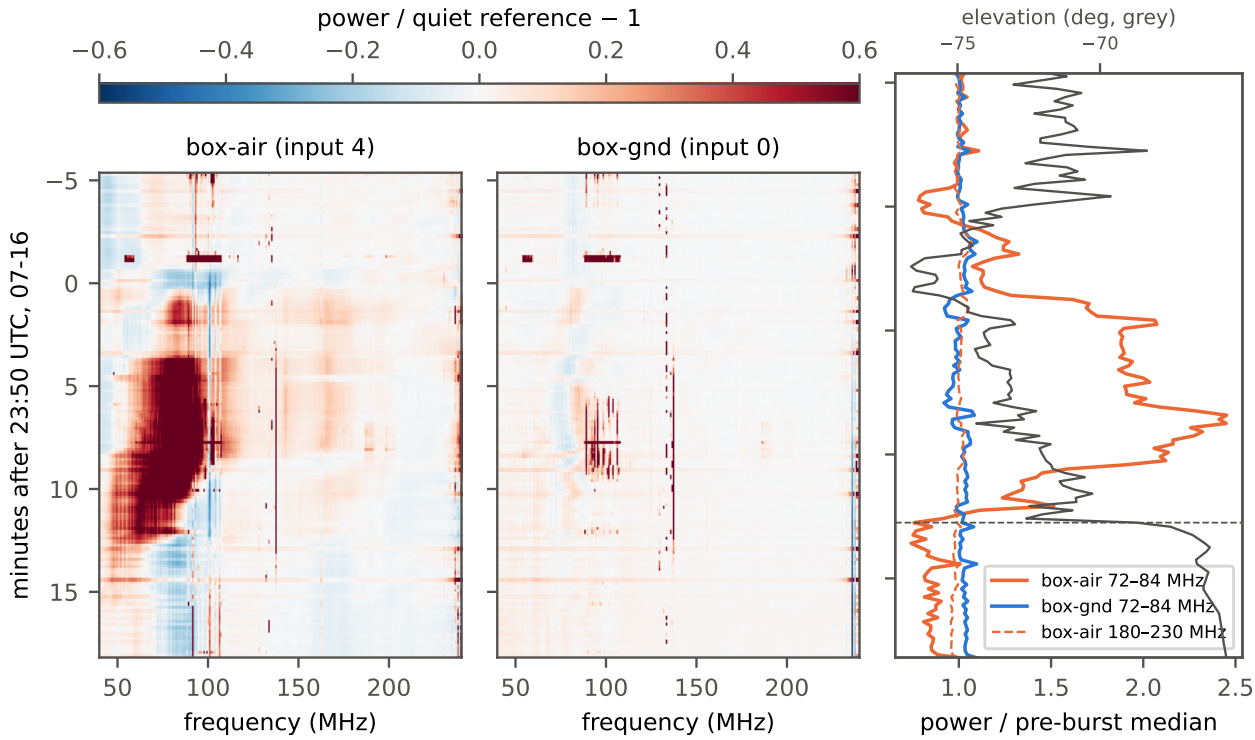


Figure 11: The 07-17 sunset in raw spectra, with box-air on the ground and tilted (`validate_sunset_burst.npz`; raw correlator files, all integrations on the antenna; 10 s means). Left and middle: box-air and box-gnd power divided by their median over quiet minutes before 23:53 and after 00:03 UTC, minus 1, with time running down. Right: band power over its pre-burst median for box-air at 72–84 MHz (solid orange) and 180–230 MHz (dashed), and box-gnd at 72–84 MHz (blue); the antenna's elevation in grey (top axis). Grey dashed: the elevation steps away from its park value. The excess is box-air's alone, drifts downward in frequency, and is mostly gone before the elevation steps.

crossing	status	fit (min)	τ^* (min)	ground (min)	121	148	199	219
					$S_\odot$ (SFU, range from T_{sys} stand-in)			
set 07-16 00:01 (30m)	fitted	-44 to +26	-4	-34	5–8	4–7	8–11	7–13
rise 07-16 16:24 (30m)	fitted; self-EMI on	-42 to +10	-16	+19	1–1†	19–32	3–5†	6–10
rise 07-17 15:54 (87.5m)	fitted	-44 to +38	-4	+50	3–5	3–4	11–16	1–2†
set 07-18 01:21 (91m)	fitted	-44 to +10	-60	-114	8–12	3–5†	7–10	4–7

Table 9: Raw-power fits at the terrain crossings (`transitions_v0005`): status, the fitted stretch and best timing shift τ^* (minutes from the crossing), when the ground under the antenna crosses ("ground", minutes), and $S_\odot$ at four frequencies (MHz) for the 5th–95th percentile of the 07-17 calibrated temperatures. †: below 2σ.

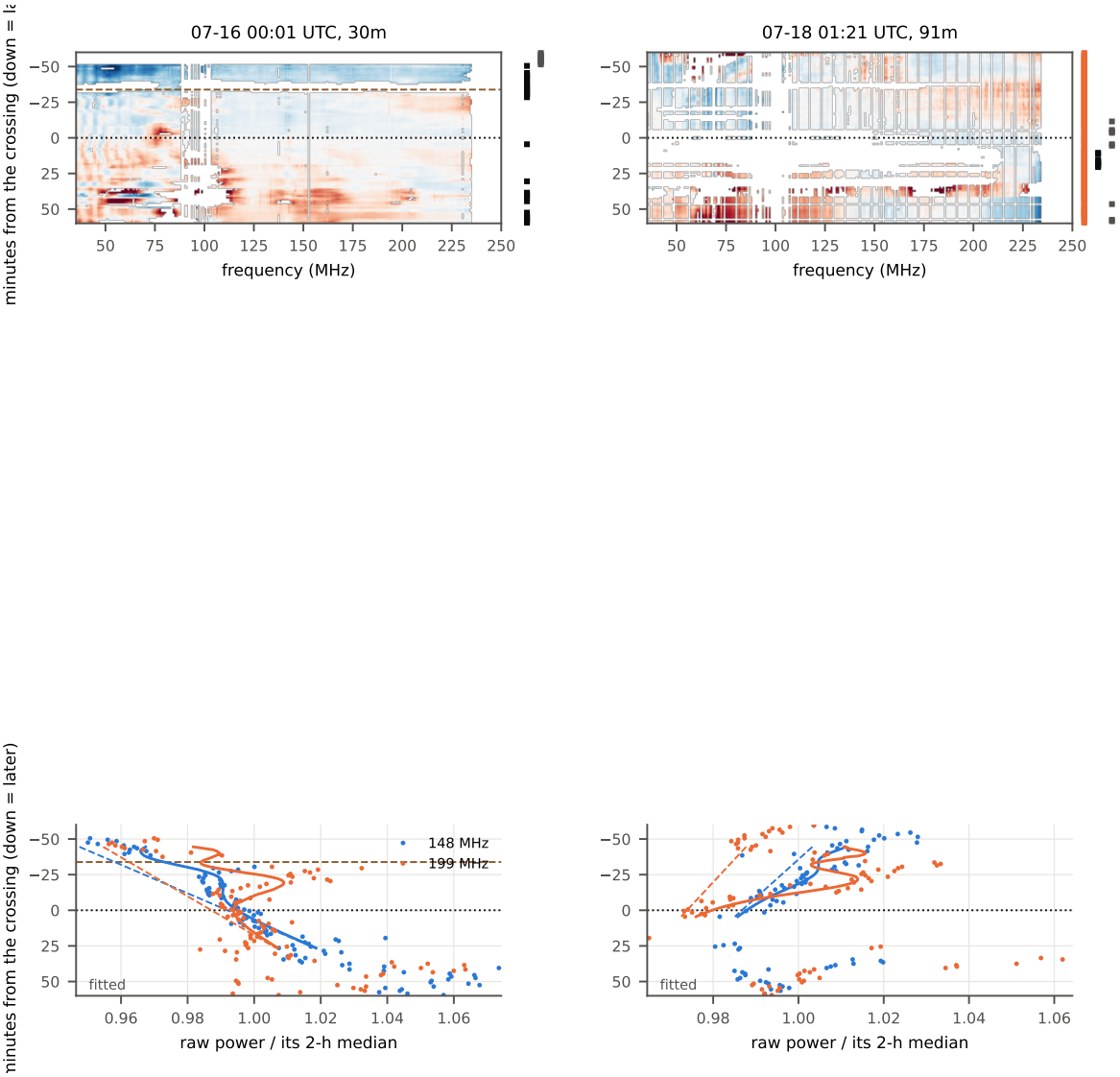


Figure 12: The three terrain sunsets in raw box-air data, one column each, time running down; the dotted line is the Sun's centre crossing the ridge at the antenna. Top: the raw spectrum, minute by minute, divided by its median over the two hours. Ticks at the right mark minutes with the transmitter on (orange), the antenna moving (black) or calibration rows (grey); the brown dashed line is when the ground under the antenna goes into shadow. Bottom: raw power at 148 and 199 MHz, divided by its median, on the same time axis. Where a fit was made, the dashed line is its baseline and the solid line baseline plus Sun. If the Sun is seen, the points follow the solid line, not the dashed one, through the half hour before the crossing.

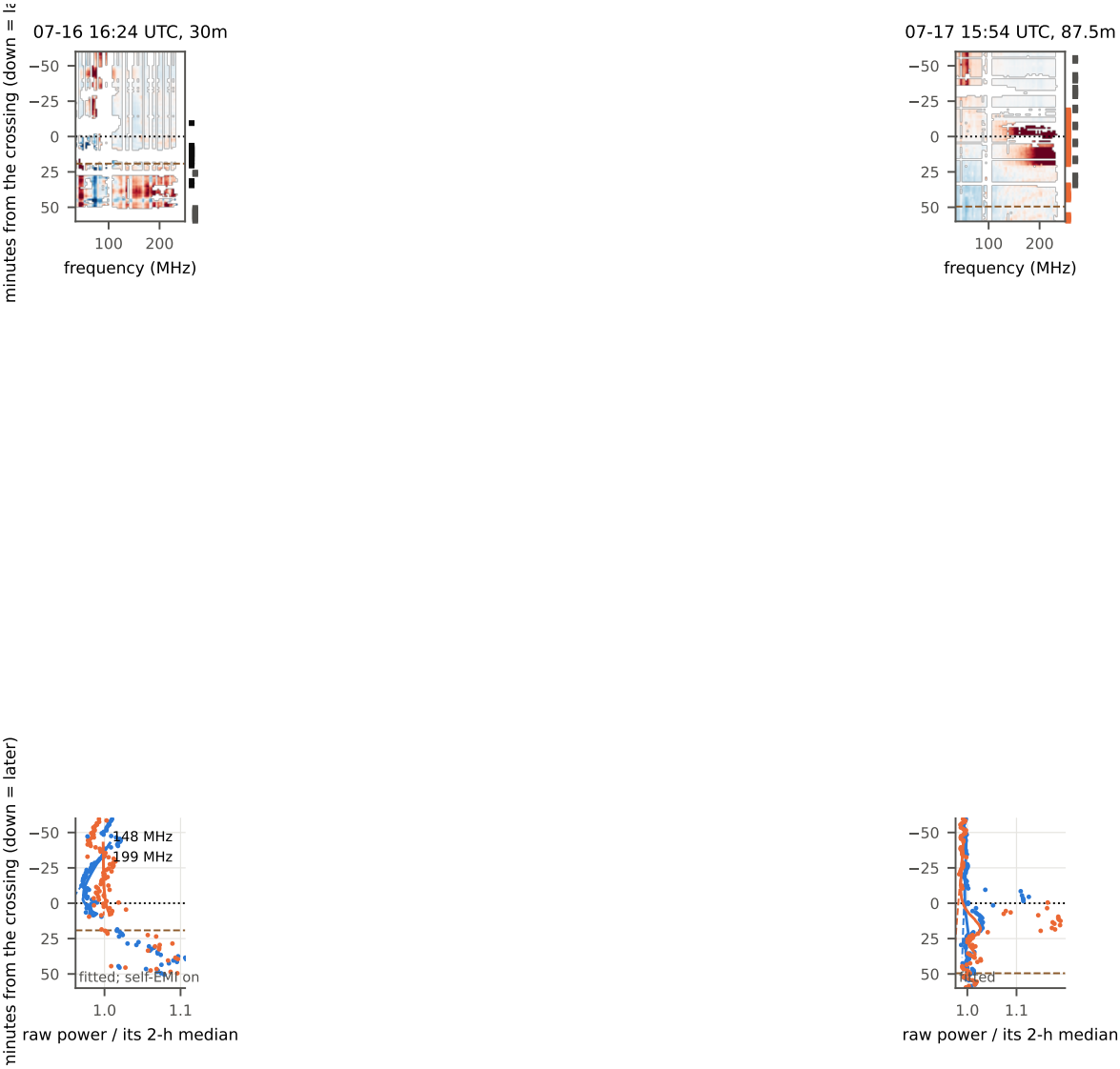


Figure 13: The three terrain sunrises, as in Fig. 12; here the brown dashed line is when the ground under the antenna comes into sunlight, and the Sun's signature follows the crossing instead of preceding it.

6.8 The Sun's path through the beam

Figure 14 shows the Sun column, minute by minute over phase C (`sun_profile_v0005`, 2320 minutes with pointing). Parked, it changes slowly with the Sun's motion, between the terrain crossings. Moving, it changes fast: during the 07-17 raster the beam's gain toward the Sun swings by factors of several within seconds. On 07-15 and 07-16 much of the day has no valid pointing in the table, and outside the raster the table's azimuth zero inherits potentiometer slips (memo 003). So the predicted Sun there is uncertain.

The raster offers a test with no kelvin scale. Since the receiver gain multiplies everything, regressing raw power on the sky (GSM through the beam), the ground, a constant, the transmitter's leakage, a drift and the Sun column gives the Sun's flux on GSM's brightness scale (`raster_sun.py`). The null is the same fit with fake Suns at the same altitude in 11 other azimuths. The test fails (Table 10). The real Sun beats every fake at 4 of 13 frequencies with the HFSS beam and 10 of 25 with the empirical beam, and the fitted fluxes change sign (7 and 20 negative). The residual, 1.9% of the power with HFSS, depends on direction and is larger than the Sun's modulation. Any extra direction-dependent column absorbs it.

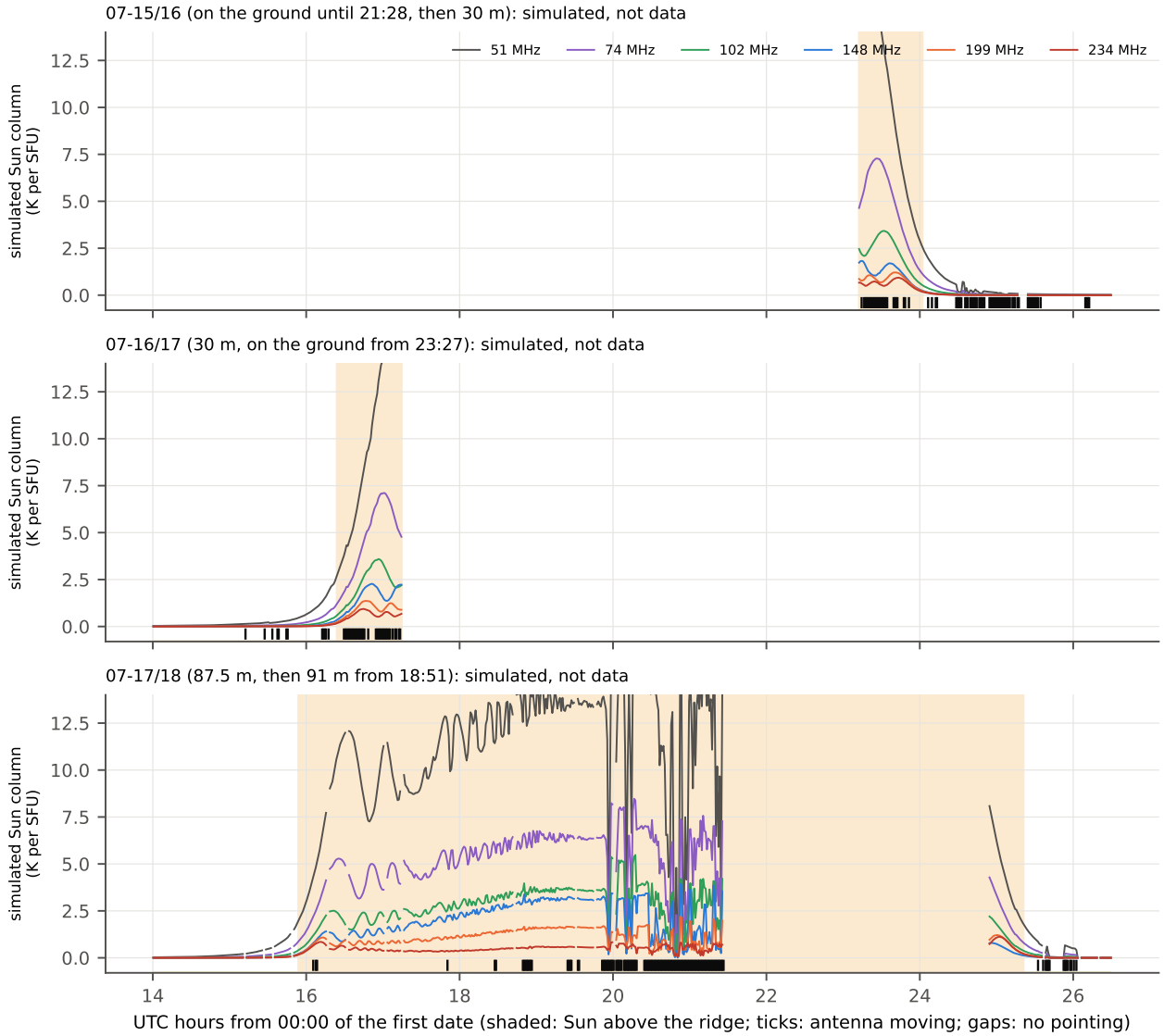


Figure 14: Simulated, not data: the predicted Sun column (antenna temperature per SFU of solar flux) through box-air's HFSS beam over phase C, at six frequencies, from the Sun's position, the antenna's recorded pointing and the knife-edge terrain visibility (`sun_profile_v0005`). No measured power is plotted.

	HFSS beam				empirical beam	
MHz	$S_\odot$ (SFU)	fakes $\geq$	resid.	MHz	$S_\odot$ (SFU)	fakes $\geq$
62.5	-3	3/11	1.3%	64.5	-0	10/11
74.2	-2	7/11	1.2%	70.3	-4	2/11
113.3	-11	2/11	1.9%	76.2	-1	8/11
125.0	-8	5/11	1.7%	82.0	-5	2/11
136.7	-19	5/11	2.0%	113.3	-12	2/11
148.4	-125	3/11	1.6%	119.1	-9	0/11
160.2	12	5/11	1.8%	125.0	-8	7/11
171.9	21	3/11	2.5%	132.8	-14	3/11
183.6	38	0/11	2.1%	138.7	-19	5/11
195.3	-40	0/11	1.8%	144.5	-47	5/11
207.0	14	0/11	2.6%	150.4	-74	3/11
218.8	95627	0/11	4.3%	156.2	106	4/11
230.5	14	1/11	8.8%	162.1	9	6/11
				168.0	39	1/11
				173.8	16	5/11
				179.7	-117	0/11
				185.5	85	0/11
				191.4	-150	0/11
				197.3	-40	0/11
				203.1	-17	0/11
				209.0	-79	0/11
				214.8	-42	0/11
				220.7	-14	0/11
				226.6	-27	0/11
				232.4	-7	5/11

Table 10: The raster test (`raster_sun_v0005`): the Sun's flux on GSM's scale, how many of the 11 fake Suns improve χ^2 as much or more, and the residual rms as a fraction of the power.

6.9 What this does and does not show

- **Box-air receives solar emission.** Solar radio bursts recorded by independent spectrographs appear in raw box-air data at the same seconds, with the type III drift and the low-frequency-weighted spectrum those stations saw (§6.5). That is a physical identification, not a coincidence in timing.
- **The raw crossings allow a quiet Sun of the expected size but do not detect it.** Across 3 uncalibrated terrain crossings on three days and three heights, the Sun term is 3–16 SFU at 121–219 MHz, against a concurrent prediction of 2–11 SFU (§6.7). It is a few-percent bump on raw power that drifts by as much for other reasons, and its kelvin stand-in is good only to a factor of 2.
- **The calibrated 07-17 excess is larger than the Sun.** At 15:54 the calibrated spectrum jumps by 3.6 times the concurrent prediction (1.4 times scaled by a), and the local fit gives 2.0–3.2 times the other crossings at 199 MHz. RSTN shows no brightening that day. Part of that excess is probably not the Sun. Anything that changes as the antenna and its box come into sunlight follows the same geometry, so it is not separated. The excess rests on 7 one-minute bins at the end of the calibrated window, all directly after the transmitter switches off.
- **The timing does not separate the Sun from the lit ground.** Best timing shifts range from -60 to -4 min, with broad minima, and 1 sits at the edge of the ± 60 min scan.
- **The beam's sweep during the raster does not isolate the Sun** with the present sky/ground/beam model: fake Suns fit as well (§6.8).
- $S_\odot$ inherits the unresolved factor $a \approx 2$ and HFSS's gain at $\sim 50°$ from boresight.

7 Recovering the sky and the ground

The night fit (§5) holds one pointing at one height, so the ground temperature and the offset are degenerate there. This section tries three ways round that, each with a check that could fail.

7.1 Tying the frequencies together

At one pointing and height, the beam's ground fraction $f_{\mathrm{gnd}}(\nu)$ still changes with frequency. If the ground is one temperature at all frequencies (thermal emission) and the receiver offset is a smooth function of frequency,

the pedestal $f_{\rm gnd}(\nu)T_{\rm gnd} + T_{\rm off}(\nu)$ could separate them. `joint_fit.py` reduces each frequency of the night fit to a likelihood for $(T_{\rm gnd}, T_{\rm off})$, marginalizing the sky and the GSM amplitude. It then fits one $T_{\rm gnd}$ and a Legendre polynomial in $\log\nu$ for the offset over 105–235 MHz. If the data fixed $T_{\rm gnd}$, it would not move with the polynomial's order, nor between the halves of the band. It does: 275 to 920 K as the order goes from 1 to 6, and the two halves disagree by hundreds of kelvin (Table 11). The ground fraction's frequency dependence is too much like a smooth offset; smoothness alone does not fix $T_{\rm gnd}$.

offset order	$T_{\rm gnd}$ (K), all	lower half	upper half
1	438 ± 59	529 ± 239	-173 ± 249
2	400 ± 132	457 ± 247	-193 ± 250
3	275 ± 136	533 ± 252	899 ± 282
4	641 ± 176	566 ± 253	927 ± 282
5	920 ± 187	566 ± 253	933 ± 282
6	820 ± 188	567 ± 253	937 ± 282

Table 11: $T_{\rm gnd}$ from the night fit tied across frequency (`joint_v0005`), against the order of the offset's polynomial, over the whole band and its two halves.

7.2 The sky as calibrator on parked nights

Most nights have no kelvin scale, but the Galaxy drifts through the beam. In a stretch of fixed pointing, raw power at each frequency is fitted as gain × (GSM through the beam) + a constant + a linear drift (`sky_cal.py`). The gain is then in raw units per GSM kelvin. The constant over the gain is $T_{\rm rx} + f_{\rm gnd}T_{\rm gnd}$ in GSM kelvin, and stretches at different heights, with different $f_{\rm gnd}$, would separate the two. Each night is fitted at one height, on its rows at that height only (§4.1); the receiver regime is rx-A throughout. A stretch enters that comparison only if the sky fits it, with a median residual under 5% of the power. Figure 15 shows the raw data with the fit.

- **07-17, 87.5 m** (calibrated): the sky fits to 0.5% rms. The gain against `tcal`'s gives a =1.8–2.1 at 105–220 MHz, as the night fit found. This checks the pipeline, not where a comes from.
- **07-16, 30 m**: box-air's self-EMI is on all night; with its teeth masked the sky fits to 3.8%.
- **07-15**: the antenna was on the ground all night, not at 30 m, and there is no horizon profile at 2 m, so the night is not fitted. Earlier versions fitted it as 30 m from 05:12 UTC; that fit is withdrawn. It also explains why its raw power before about 04:45 was a few percent of its later level.

That gives 2 nights, whose ground fractions span only 0.05. A 10 K difference between the nights in anything else then shifts $T_{\rm gnd}$ by about 188 K, and the fit gives −2642 to 2044 K at 120–180 MHz. Parked nights at these two heights cannot separate the ground.

7.3 The ground from the rotating antenna

When the antenna moves, the beam's ground fraction swings between about 0.5 and 0.75 within seconds, while the receiver gain stays put. In four moving windows (`raster_sky.py`), raw power per integration is regressed on GSM through the beam, the ground fraction, a constant, the transmitter's leakage, the Sun and a drift. The gain multiplies every column, so the ground coefficient over the GSM coefficient is $T_{\rm gnd}$ in GSM kelvin, with no kelvin calibration. Errors are jackknifed over 2-minute blocks. Each window uses only its rows at the window's height (§4.1). The 07-18 night window keeps the 8290 of its 16260 rows after the re-raise; 4370 were on the ground and 3600 of unknown height. The raster loses its last 240 rows, of unknown height. Earlier versions also used a 30 m dusk window on 07-17 00:21–01:08, which was on the ground; it is dropped.

The excess in the raw data. Figure 16 shows it in the raster's raw power, integration by integration, as the beam's ground fraction swings between 0.60 and 0.75 over each elevation sweep. Take the raw power with no RFI flags at all, and subtract every fitted term except the ground: the sky, the constant, the transmitter, the Sun and the drift. What is left at 64 and 82 MHz rises and falls with the ground fraction by 31 and 43% of the power, peak to peak. A 300 K ground would move it by a few percent. The slope of that residual against the ground fraction is the ground's brightness: 2309 and 1637 K in GSM kelvin with no flags, and 3620 and 1797 K

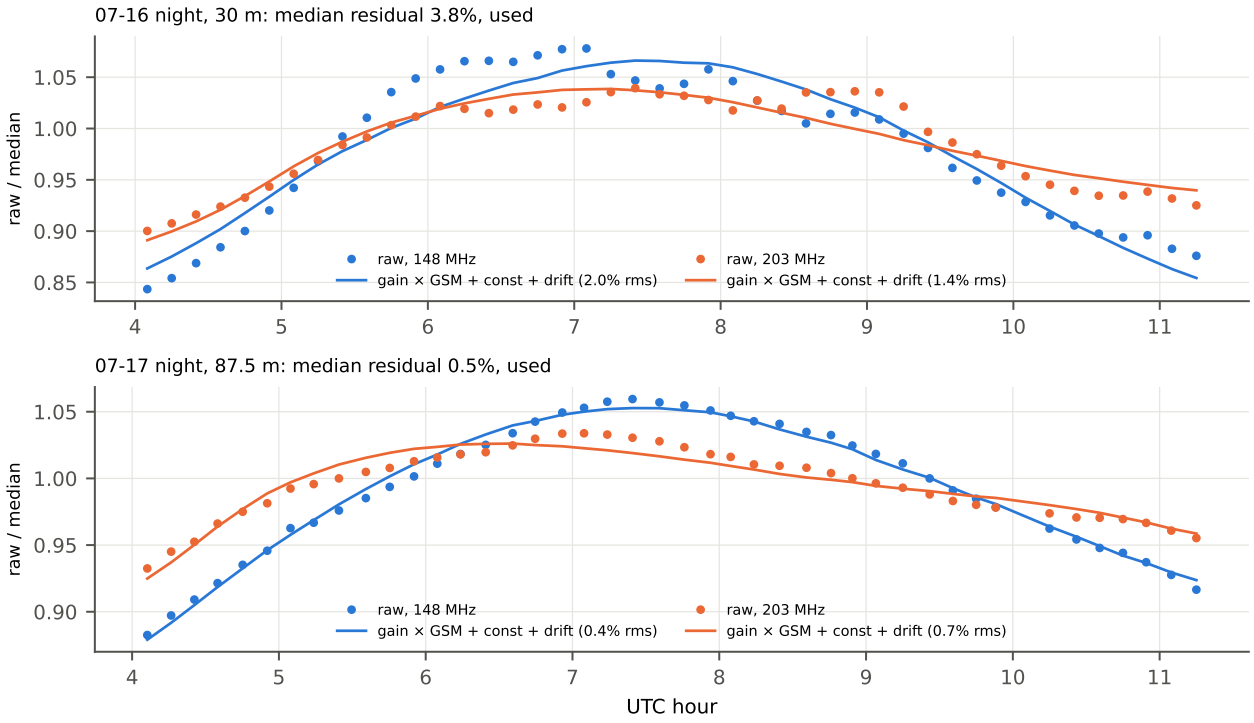


Figure 15: The parked nights at a confirmed height in raw data at 148 and 203 MHz, each divided by its median (points), with the fitted gain × GSM through the beam + constant + drift (lines) (`skycal_v0006`, HFSS beam). The 07-15 night, on the ground, is not fitted.

on the RFI-flagged samples (19 and 22% of them). So at low frequency the excess is in the raw data, and the flags change its size but did not create it. At 162 MHz it is small and depends on the flags: 120 K unflagged and 236 K flagged (53% kept). At 113 MHz the unflagged raster is dominated by the transmitter's leakage, so only flagged values are quoted there. The flags keep fewer than a third of the raster's samples below 120 MHz, and the fraction kept depends on where the beam points. The flagged values above 100 MHz therefore carry a selection that the unflagged ones do not.

Figure 17 and Table 12 show the result in every window.

- **It repeats.** The three 91 m windows (two evenings, the transmitter and the Sun in different states) agree at 6 frequencies between 120 and 180 MHz (χ^2/dof up to 12.5).
- **It is not flat in frequency.** It falls from 466 K at 125 MHz and 346 K at 139 MHz to 196–246 K at 150–175 MHz. Below 85 MHz every window gives thousands of kelvin: 1970–3620 K at 64 MHz.
- **The 30 m window sits lower and scatters more**; it has the self-EMI on.

Thermal ground emission should not depend on frequency. At 150–175 MHz the values are those of soil at about 290 K with an emissivity near 0.7, but that coincidence does not identify them. What rises below 140 MHz is not the ground's temperature. It enters wherever the beam sees the ground, and it grows steeply toward low frequency like the sky.

Not reflected sky. Ground reflects part of the sky, which is far brighter at low frequency. A specularly reflected sky term (each below-horizon direction mirrored to the sky above, flat ground) was added in the raster, with the ground's reflectivity R free. Soil would give $R \approx 0.1$–0.4. At 64–82 MHz the fit gives $R = -0.15$ to -0.03, and the ground's emission term stays at 1831–3704 K. Specular reflection does not explain the excess, and diffuse reflection is tested over phase C in §7.5. Candidates left are the beam below the horizon (both beams are extrapolated there; the raster does not sample it, memo 005) and something fixed to the platform that turns with the antenna.

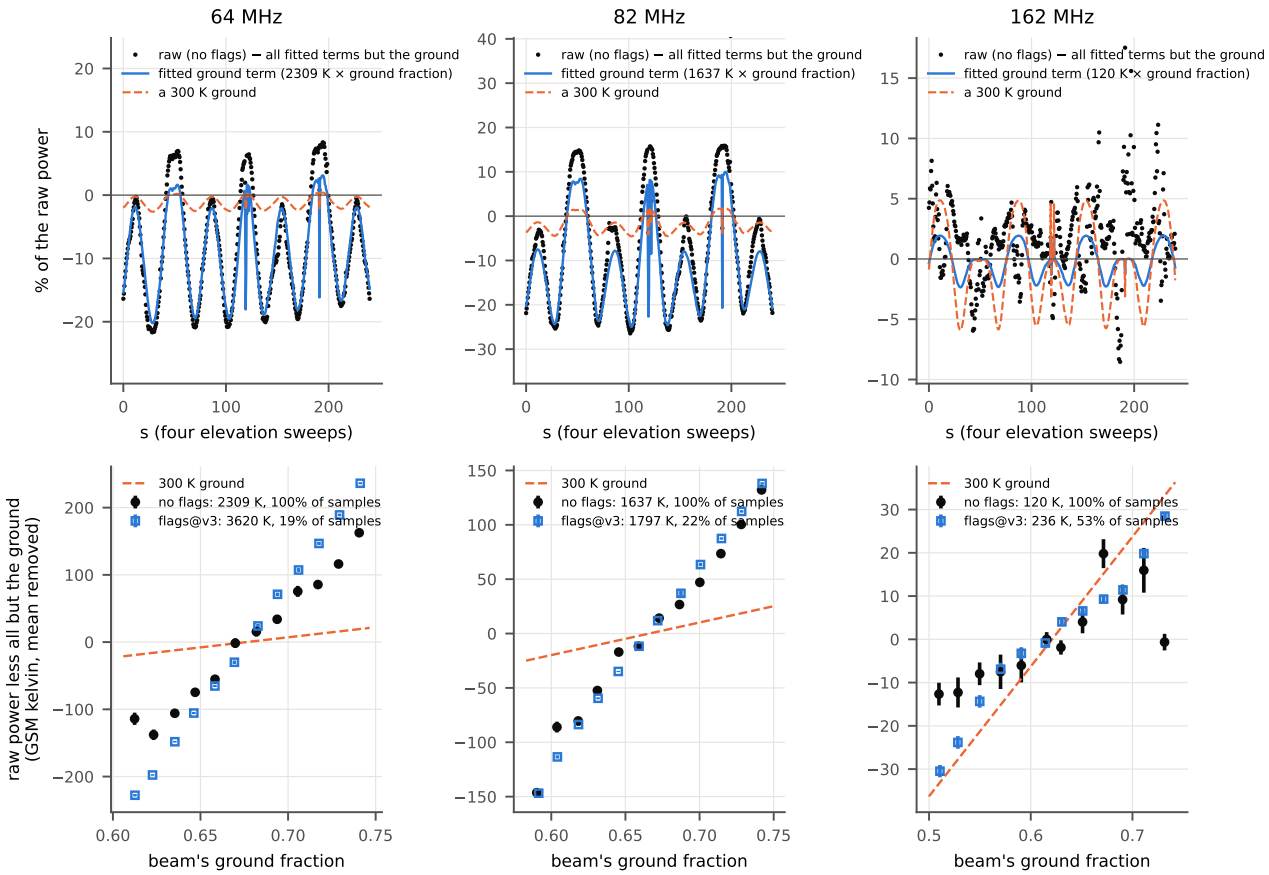


Figure 16: The ground excess in the raster's raw power (`raster_sun_v0005`, raw correlator files for the unflagged values; channels midway between the transmitter's comb teeth). Top: four elevation sweeps of raw power with no RFI flags, less every fitted term except the ground (points), against the fitted ground term (blue) and what a 300 K ground would give (orange), in percent of the power. Bottom: the same residual, binned by the beam's ground fraction and converted to GSM kelvin, with no flags (filled) and with `flags@v3` (open); the slope is the ground's brightness.

MHz	mean of three 91 m windows (GSM K)	χ^2/dof
125.0	466 ± 40	2.2/1
132.8	346 ± 25	0.8/1
138.7	364 ± 26	25.0/2
162.1	246 ± 18	0.1/1
168.0	217 ± 14	0.7/1
173.8	196 ± 14	0.1/1

Table 12: The ground term, weighted mean of the three 91 m windows, where at least two have errors under 200 K, and the windows' agreement.

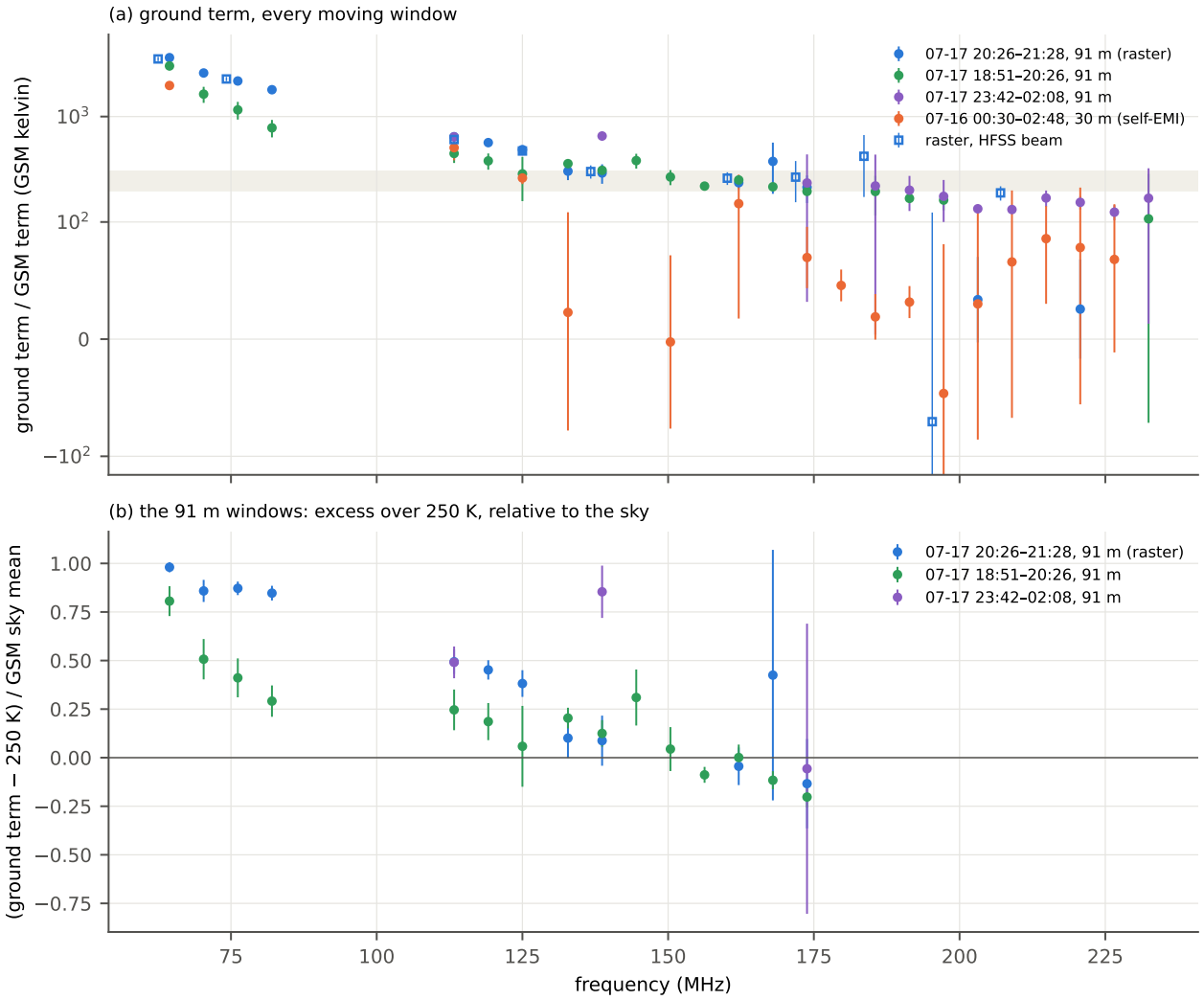


Figure 17: (a) The ground term in each moving window, as the ratio of the ground and GSM coefficients (GSM kelvin; empirical beam unless marked; jackknife errors; points with errors above 300 K omitted). Shaded: 200–300 K. The axis is linear below 100 K and logarithmic above. (b) For the three 91 m windows, the same ground term less a thermal 250 K, divided by GSM's whole-sky mean (below 180 MHz). A sky leaking in with a fixed fraction would give a constant.

7.4 Tracing the excess

Four explanations of the excess make different predictions, and the rotating-antenna data can test three of them.

Where on the ground. `ground_excess.py` repeats the regression with the ground split into regions, each with its own brightness (Table 13).

- **Horizon band against deeper ground.** Radio noise from the landscape, man-made or lightning, would arrive from just below the horizon. In the raster the band within 10° below the horizon and the deeper ground carry similar excesses with the flags. With no flags the band is somewhat brighter, but the deep ground still carries thousands of kelvin at 64–84 MHz. The deep ground, most of the beam's ground weight, is not reached by noise from the horizon.
- **Azimuth.** Sources on the landscape would sit at particular azimuths. In the raster, eight azimuth sectors give 1767–2542 K at 76 MHz (typical error 311 K): the excess covers the ground evenly. The other windows sample too few orientations to constrain eight sectors.

Against the sky. A sky seen where the model expects ground would give an excess proportional to the sky's brightness. That covers sky scattered diffusely by rough ground, or a back lobe stronger than modelled. Figure 17b divides the excess (the ground term less 250 K) by GSM's whole-sky mean. In the raster the ratio is 0.85–0.98 at 64–82 MHz and falls to near zero by 150–175 MHz (within ±0.4). It also differs between windows. Its spectral index between 64 and 82 MHz is 3.2, steeper than the sky's (≈2.5). A constant scattering fraction or back-lobe gain does not reproduce this; a coupling to the sky that weakens fast with frequency would.

A gain that follows the orientation. If the system gain or the antenna's match changes as the antenna turns, by an amount that follows the ground fraction, the change multiplies a nearly constant system power. It then has the same shape as the ground column and cannot be told apart from it by these regressions. A gain swing as large as the ground term's swing over a sweep (Fig. 16: 31% of the power at 64 MHz, 43% at 82 MHz, 5% at 162 MHz, peak to peak) would make the whole excess. A gain change that multiplies only the sky (an extra column of ground fraction times GSM) does not absorb it: at 64–82 MHz the ground term stays at 2372–3712 K, against 1797–3620 K without it. Testing an orientation-dependent match needs the antenna's reflection coefficient at different tilts. All 19 field antenna-S11 sweeps from 07-13 on were taken upright: those with pointing (19) are within 3.2° of el = 0. So the data cannot test it.

So far. The excess is not specular reflection of the sky, not noise arriving from the horizon, and not concentrated in any azimuth. It does not scale with the sky as a fixed fraction. What is left is a coupling that falls steeply with frequency and follows the antenna's orientation relative to the ground: either the beam's response below the horizon, which both beam models extrapolate, or the antenna's match or gain changing as it tilts. Telling them apart needs a measurement, not more regression: S11 or a calibration cycle with the antenna tilted, or a beam measurement that samples directions below the horizon. The daily term over phase C (§7.5) is a separate test, with the sky drifting under fixed pointings.

window, region	64	70	76	82	113	125
raster, flags@v3: within 10° below horizon	4175±236	3019±496	2379±261	1637±239	970±140	920±340
raster, flags@v3: deeper ground	3574±95	2613±169	2179±83	1793±73	653±48	510±51
raster, no flags: within 10° below horizon	5330±1120	3152±1144	3349±635	2646±423	–	315±356
raster, no flags: deeper ground	2579±851	1830±160	1777±269	1651±135	24±801	391±47
07-17/18 night, flags@v3: within 10° below horizon	–	–	–	–	–	–
07-17/18 night, flags@v3: deeper ground	–	2619±494	3591±1278	–	629±55	–
raster, 64 MHz, 8 azimuth sectors	4079±957, 3070±725, 5100±1432, 4422±1588, 3279±527, 5644±1089, 2061±599, 4008±549					
raster, 76 MHz, 8 azimuth sectors	2300±272, 2180±329, 2177±316, 1767±232, 2542±337, 1810±236, 2312±315, 2323±306					

Table 13: The ground split into regions (`ground_excess_v0005`, empirical beam; brightness in GSM kelvin, jackknife errors; "–": error above 2000 K). Top: the band within 10° below the terrain horizon and the deeper ground, at six frequencies (MHz). Bottom: the raster's eight azimuth sectors (0–45°, 45–90°, . . . , east of north).

7.5 The low-frequency daily term over phase C

The moving windows each last about an hour, so the sky hardly changes within one and cannot be told from a fixed ground term. Phase C adds what they lack: three days of sidereal time. `global_fit.py` models rx-A raw power in `raw_binned_v0002` over all of phase C (4236 bins in 36 gain windows that pass its residual gate), one frequency at a time. The model is a smooth gain times [GSM through the HFSS beam + a ground term + an offset]. The log gain is piecewise linear in time, with knots every 3 h and a 5% prior on each knot step. There are no sky corrections ($l = 0$). Alternate 30 min blocks are held out, and every model is scored by the held-out rms as a fraction of the power. The code is in data-analysis, run at commit 79e9960. `validate_phase_c.py` in this directory runs each configuration on the 12 beam frequencies of the original analysis, 62.5–175.8 MHz; the 6 from 62.5 to 82.0 MHz are reported. The ground term takes one of these forms, each with one free parameter unless stated:

- a constant temperature $T_{\rm g}$;
- diffuse (Lambertian) reflection of the sky, $(1 - R)\,290\,{\rm K} + R\,\bar{S}(t)$. Here $\bar{S}$ is the cosine-weighted GSM mean over the open sky above the era's horizon, 1153–5184 K at these frequencies, so a ground term of thousands of kelvin is the size this form would give;
- the same, with $\bar{S}$ taken 6 h earlier, 6 h later or 12 h away in sidereal time, as controls;
- "beam": the beam's response below the horizon falling on the sky of the current pointing instead of on the ground;
- "pickup": $k\,\bar{S}$ without the ground fraction, for sky entering by a path that does not turn with the antenna;
- a housekeeping stream, averaged over each bin from the correlator metadata: three RF-switch thermistors, the load temperature and the system current;
- "free level": $T_0 + R\,\bar{S}$ with both free and a 290±300 K prior on T_0.

Heights. Phase C here is only its rows at 30, 87.5 and 91 m (§4.1). Earlier versions also fitted the 07-15 night and the 07-17 sunset park as 30 m, which were on the ground; most of the term was found with those rows in, and the results below differ from those versions accordingly.

Over all of phase C (Table 15), reflection with the real sky beats a constant ground by 0.20–0.51 points of held-out rms, with $R = 0.41$–1.04. The shifted skies gain up to 0.50 points, nearly as much, so the real sky is not singled out. The beam form differs from a constant ground by at most 0.08 points, so the term does not follow the sky of the current pointing. Pickup does better than reflection (by 0.22–0.30 points, with 2 of its fits diverged), so the ground fraction is not what carries the term.

Against recorded temperatures and current. On the 100% of bins with all five streams, the best single stream does between −0.07 and 0.34 points worse than reflection at 62–78 MHz (a negative value: better); at 82 MHz the first thermistor reaches 3.17% against 2.98% for reflection. With all five streams in the fit at once, reflection with the real sky still lowers χ^2/dof by 4.5–8.4, against at most 0.7 for the shifted skies, and R stays 0.44–0.71 at 62–74 MHz. So χ^2 still prefers the real sky beside the streams, while the held-out residual does not separate it from the shifted skies.

With the level free the ground's constant part is no longer tied to R. The sky shifted by 6 h then does worse than the tied real sky at 62–70 MHz (by −1.03 to −0.63 points, negative meaning worse), and at 74–82 MHz the tied real sky beats every shifted free-level fit by 0.16–0.45 points. The free real sky itself (0 of 6 fits diverged) gives $R = 3.5$–5.2 with T_0 from −10 to −7 kK: far more time variation than reflection allows, offset by a negative level.

Disjoint subsets (Table 16). Splitting phase C at 07-16 18:10 UTC gives halves A and B. In half A the reflection form is the best model at 6 of 6 frequencies, and the 6 h-shifted sky does 0.3–0.9 points worse; χ^2/dof is 50–70. Half B (1424 bins) fits at χ^2/dof 6–9, and reflection gains −0.10 to 0.00 points over a constant ground there: no term. The look-down park that dominated half B's misfit in earlier versions was the antenna tilted on the ground, and is now out. Half A split by day at 07-16 00:00 UTC gives 07-15 (311 bins, only the evening

at 30 m), where R is unconstrained, and 07-16 (2501 bins), where R is −0.06 to 0.17 (errors up to 0.06) and a shifted sky is the best model at 3 of 6 frequencies. Neither day alone separates the real sky as half A does.

Window selection. `global_fit.py` drops gain windows whose median residual over the fitted frequencies is above 5% of the power, so which windows enter can depend on the frequencies fitted. Gated on 62–82 MHz alone, the same 4236 bins in 36 windows pass as with all 12 frequencies, so the gate does not change these results.

Faster gain. The fits above put a gain knot every 3 h, and §7.6 finds the gain changing within the hour. `validate_phase_c_fastgain.py` refits phase C with knots every 3 h, 1 h and 20 min, and injects that sub-hour gain into mocks (Table 14; data-analysis commit 79e9960). Every fit anchors the GSM amplitude in tcal kelvin at $a_{\mathrm{tcal}} = 1.5$, this model's level at 62–82 MHz with 3-h knots, either softly (±0.02 in log, which these data outweigh) or hard ($\pm 10^{-4}$).

- **With faster knots the term goes.** With 3-h knots, reflection with the real sky beats a constant ground by 0.18 to 0.47 points. With 1-h knots the difference is −0.04 to 0.45, and with 20-min knots −0.41 to 0.00. Freed this way, the gain level is $a_{\mathrm{tcal}} = 1.01$–1.31 against the soft anchor at 1-h knots. Held at 1.5 by the hard anchor, reflection is fitted with $R = -0.22$–0.40 at 1-h knots and does no better than a constant ground (−0.37 to 0.03 points).
- **The gain takes up the sky's daily shape.** Per gain segment, the log gain knots of the constant-ground fits correlate with $\log \bar{S}$ of the real sky at +0.40 to +0.84 (3-h knots), +0.87 to +0.92 (1-h) and +0.86 to +0.93 (20-min), and with the shifted skies at −0.61 to +0.11 (1-h) and −0.60 to +0.13 (20-min).
- **The sub-hour gain does not make the term.** The mocks take the 3-h fits as truth, times the frequency-flat fast gain of §7.6 (1.9% rms, its temperature-driven part included). Without a daily term, reflection gains −0.02 to 0.03 points with 3-h knots, and with the real data's own residuals multiplied in −0.12 to 0.10, against the real data's 0.18–0.47.
- **The residuals would hide a real term.** A mock with a daily term ($R = 0.41$–0.93), the fast gain and the real residuals gains only −0.05 to 0.20 points with 3-h knots and −0.06 to 0.18 with 1-h knots, although R is recovered at 1-h knots (0.13–1.00). Without the residuals it gains 0.13–0.39 points with 3-h knots, 0.00–0.03 with 1-h and −0.03–0.38 with 20-min knots. With 20-min knots and the residuals, the comparison is erratic in both mocks: −0.08 to 0.18 points without a term and −0.05 to 0.28 with one. So, given the residuals these fits leave, the gain comparison cannot show a term of this size reliably at any knot spacing, and its loss with faster knots does not rule a term out.

	3 h knots		1 h knots		20 min knots	
	gain	R	gain	R	gain	R
Real data, soft anchor	+0.18 to +0.47	0.41 to 0.93	−0.04 to +0.45	−0.41 to 0.09	−0.41 to +0.00	−0.85 to 0.26
Real data, hard anchor	–	–	−0.37 to +0.03	−0.22 to 0.40	−0.55 to +0.05	−0.33 to 0.37
Mock, no daily term, fast gain	−0.02 to +0.03	0.17 to 0.41	−0.03 to +0.02	−0.04 to 0.39	−0.21 to +0.06	−0.03 to 0.43
Mock, daily term, fast gain	+0.13 to +0.39	0.51 to 0.98	+0.00 to +0.03	0.38 to 0.86	−0.03 to +0.38	0.40 to 0.87
Mock, no daily term, fast gain and residuals	−0.12 to +0.10	0.11 to 0.42	−0.07 to +0.10	−0.32 to 0.47	−0.08 to +0.18	−0.26 to 0.54
Mock, daily term, fast gain and residuals	−0.05 to +0.20	0.46 to 1.19	−0.06 to +0.18	0.13 to 1.00	−0.05 to +0.28	0.25 to 0.89
Gain knots against $\bar{S}$: real sky	+0.40 to +0.84		+0.87 to +0.92		+0.86 to +0.93	
shifted skies	−0.49 to +0.26		−0.61 to +0.11		−0.60 to +0.13	

Table 14: The daily term against faster gain (`validate_phase_c_fastgain.json`), at 62–82 MHz. "gain": held-out rms of a constant ground minus that of reflection with the real sky, in percentage points (positive: the daily term helps), over the converged fits; R: the fitted reflection coefficient. Soft and hard anchors hold a_{tcal} at 1.5 by ±0.02 and $\pm 10^{-4}$ in log. The mocks are the 3-h fits times the fast gain of §7.6, with radiometer noise, and in the last two rows with the real fits' residuals multiplied in (rolled by 1500 bins). Bottom: the correlation of the constant fits' log gain knots with $\log \bar{S}$, per gain segment.

What this shows. On rows at confirmed heights, a slow additive term at 62–82 MHz still improves the phase-C fit with 3-h gain knots over a constant ground, but the evidence that it is sky has gone. Skies shifted by 6 or 12 h in sidereal time do nearly as well, the term is absent from half B and from 07-16 alone, and the fitted gain follows the open-sky mean even with 3-h knots (+0.40 to +0.84). Earlier versions found it mostly in rows now

known to be on the ground. Where the receiver's gain after the RF switch is measured (§7.6), it does not follow the sky. The 30 m era has one usable calibration interval, 07-16 01:41–02:24 UTC with the antenna moving, in half A; over it the gain after the switch changes by at most 0.9% at 110–205 MHz, against −1.7 to 5.0% for the 20-min fit's gain, and at 43 min it is far too short to show a daily term. Whether the term is additive, a gain change on 1–3 h scales, or the fit absorbing sky or beam mismatch, these data cannot tell. Over phase C sidereal time gains only 3.9 min a day on solar time, so separating a sky term from an instrumental daily cycle needs observations weeks apart; the ground rows could add a second height once a 2 m horizon exists.

MHz	constant	real	+6 h	−6 h	12 h	beam	pickup	R (real)
62.5	4.00	3.49	4.49	3.80	3.83	3.98	**3.27**	0.65±0.03
66.4	3.59	3.12	4.06	3.44	3.43	3.66	**2.82**	0.51±0.03
70.3	3.39	3.11	3.74	3.45	3.43	3.39	**2.81**	0.42±0.03
74.2	3.44	3.24	3.60	3.35	3.34	3.45	**2.96**	0.41±0.03
78.1	4.18	3.80	4.10	3.79	**3.68**	4.24	3.46†	0.53±0.04
82.0	3.45	**2.98**	3.51	3.17	3.17	3.48	2.77†	1.04±0.08

Table 15: The phase-C fits on all bins (`validate_phase_c.json`): held-out rms in percent of the power for each ground form (one free parameter each; "real" and the shifted skies are the reflection form), and the reflection coefficient R of the real sky. Bold: the best fit in the row. †: the fit diverged (a gain knot below 1% of the median, or an offset above ten times the GSM column).

MHz	constant	real, tied	real, free	+6 h	−6 h	12 h	R (real, tied)
Half A							
62.5	4.80	4.32	**4.11**	5.21	4.88	4.81	0.58±0.04
66.4	4.25	3.81	**3.45**	4.73	4.58	4.38	0.46±0.03
70.3	3.91	3.70	**3.27**	4.19	4.17	3.91	0.37±0.03
74.2	4.30	4.20	**3.70**	4.49	4.40	4.18	0.36±0.03
78.1	5.82	5.42	**4.92**	6.02	6.05	5.66	0.45±0.05
82.0	4.91	4.38	**4.12**	5.11	4.98	4.86	0.86±0.09
Half B							
62.5	**2.44**	2.19†	2.44†	2.54	2.53	2.48	2841.98±320.39†
66.4	2.17	**2.01**	2.20	2.40	2.23	2.18	5.10±1.60
70.3	2.19	**2.17**	2.21	2.56	2.24	2.19	3.68±0.99
74.2	1.85	1.85	1.95	2.10	1.87	**1.84**	3.41±0.82
78.1	1.65	**1.65**	1.70	1.80	1.69	1.71	1.49±0.17
82.0	**1.70**	1.80	1.76	2.06	1.73	1.74	1.29±0.16
07-15							
62.5	nan†	nan†	–	nan†	nan†	nan†	-2502.69±275.11†
66.4	nan†	nan†	–	nan†	nan†	nan†	-4696.27±479.79†
70.3	nan†	nan†	–	nan†	nan†	nan†	2057.39±2521.71†
74.2	nan†	nan†	–	nan†	nan†	nan†	2228.89±19.35†
78.1	nan†	nan†	–	nan†	nan†	nan†	1.27±0.58†
82.0	nan†	nan†	–	nan†	nan†	nan†	1372.68±899.16†
07-16							
62.5	4.15	**4.00**	–	4.61	4.26	4.04	0.17±0.06
66.4	4.23	**3.82**	–	4.35	4.14	3.90	0.15±0.04
70.3	3.86	3.85	–	4.49	4.08	**3.78**	0.05±0.04
74.2	4.09	4.12	–	4.67	4.35	**3.77**	0.02±0.05
78.1	5.36	5.26	–	6.12	5.95	**5.02**	-0.06±0.06
82.0	4.63	**4.37**	–	4.98	4.87	5.26	0.11±0.06

Table 16: The same comparison on disjoint parts of phase C (`validate_phase_c.json`). Half A is phase C before 07-16 18:10 UTC, half B after it. 07-15 and 07-16 split half A at midnight UTC. "real, free" and, in the halves, the shifted skies have a free level with the 290±300 K prior; in the two days the shifted skies are tied, like the real sky. Bold: the best converged fit in each row; †: diverged.

7.6 The ground term with a fast gain

The phase-C fits above hold the gain smooth over 3 h. Freeing it on shorter timescales changes the ground term several-fold, so this subsection asks which gain the data support and what ground term follows. `global_fit.py` (data-analysis commit 79e9960) fits rx-A in `raw_binned_v0002` over phase C at 21 frequencies from 51 to 234 MHz, one at a time: a smooth gain times [GSM through the HFSS beam + T_{gnd} times the ground fraction

+ an offset + the transmitter's leakage + the Sun], with sky corrections to $l = 2$. The gain prior allows 10% per knot step. Bins during the 07-16 23:50–00:02 UTC solar flare (§7.5) are excluded, which leaves 4236 bins in 36 windows. Alternate 5 min blocks are held out. `validate_ground_term.py` in this directory runs every fit quoted here.

Which gain. With knots every 20 min instead of 3 h, $T_{\rm gnd}$ at 117–180 MHz rises 1.7–4.5-fold (198 K to 333 K at 148 MHz; at 1 frequency the 3-h value is negative), while at 203 MHz the two agree (226 and 199 K). The held-out rms falls by 0.3–0.7 points. Held-out blocks score how well each model predicts nearby bins, not whether its $T_{\rm gnd}$ is right, so the choice is tested by injection (Fig. 18). A mock is one fit's own prediction for every bin, with radiometer noise, times the 20-min fit's real residuals rolled in time, so they keep their size and timescale but not their alignment with pointing. Its $T_{\rm gnd}$ and gain are then known.

- A mock with the 3-h fit's slow gain is recovered by both knot spacings, the 20-min one to 91 K rms (at most 130 K; 114 K at 148 MHz). Freeing the gain does not by itself inflate $T_{\rm gnd}$.
- A mock with the 20-min fit's gain, refitted with 3-h knots, returns the low values the real data give with 3-h knots, within 110 K at every frequency (169 K at 148 MHz).

The real data behave like the second case: the model needs a multiplicative term that changes within the hour, and slow knots leave those changes to be absorbed by the ground and offset terms. Below, it is called the fast gain; where the receiver's own gain is measured it is not that ("The receiver's own gain"). The 20-min fit is the one used below.

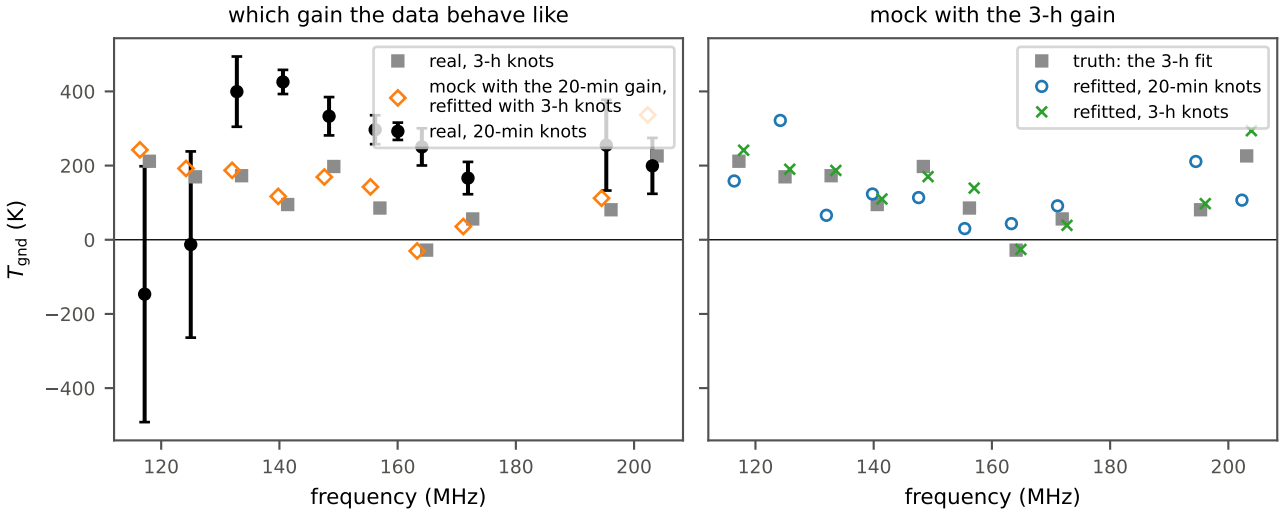


Figure 18: $T_{\rm gnd}$ at 110–205 MHz (rx-A, `raw_binned_v0002`, `validate_ground_term.json`; frequencies where the 20-min fit collapses omitted). Left: the real data with 20-min gain knots (errors from the injections) and 3-h knots, and a mock with the 20-min fit's gain refitted with 3-h knots, which reproduces the real 3-h values. Right: a mock with the 3-h fit's gain (its truth in grey), recovered by either knot spacing. Points are offset by 0.8 MHz for legibility.

The ground term and its error (Table 17). At 110–205 MHz $T_{\rm gnd}$ falls from 425 K to 199 K. The error is the rms over 3 injections of recovered minus true $T_{\rm gnd}$ (two rolls of the residuals on the 20-min mock, one on the 3-h mock): 33–122 K, 0.9–2.6 times the fit's own error (23–46 K), which takes the residuals as white. The draws average −2 K. With three draws per frequency the error is a scale, not a precise σ. The residuals in the mocks do not follow the pointing. Real residuals that do, from terrain or beam errors, could bias $T_{\rm gnd}$ in a way these injections do not test. This ground term and the rotating antenna's (Table 12) both fall with frequency over 125–175 MHz, but they come from different models, beams and data, and are not combined.

Where the fit collapses. At 180, 188, 219 MHz the 20-min fit has two solutions of nearly equal χ^2. In one, $a_{\rm tcal}$ is near its values elsewhere in the band (1.1–73.1). In the other the gain collapses, to $a_{\rm tcal}$ =0.003–0.49 (at

MHz	T_{gnd} (K)	injection (K)	fit (K)	a_{tcal}	
50.8	10468	3477	3488	3.57	unstable
58.6	3357	153	168	7.89	
66.4	1877	584	60	5.04	
74.2	1063	291	32	5.28	
82.0	-44	930	1308	1.55	unstable
117.2	-147	345	297	1.12	unstable
125.0	-13	251	93	1.96	unstable
132.8	399	95	39	4.42	
140.6	425	33	37	4.06	
148.4	333	52	34	3.66	
156.2	297	39	25	4.67	
164.1	250	50	23	5.01	
171.9	166	44	57	2.20	unstable
179.7	$105^{\ddagger}$	44	–	$2.40^{\ddagger}$	shared-gain fit
187.5	$97^{\ddagger}$	20	–	$2.90^{\ddagger}$	shared-gain fit
195.3	255	122	46	4.55	
203.1	199	75	42	5.15	
210.9	20	nan	300	1.11	
218.8	–	–	–	0.003	collapsed
226.6	76	16	13	73.13	unstable
234.4	467	133	129	29.06	unstable

Table 17: T_{gnd} over phase C with 20-min gain knots (rx-A, `validate_ground_term.json`). "injection": the rms of recovered minus true T_{gnd} over the injections, the error to use; "fit": the fit's own error. a_{tcal} is the GSM amplitude, gain times `tcal`'s scale. "collapsed": a_{tcal} below 0.50 (or diverged), where no T_{gnd} is quoted. "unstable": collapses once the transmitter's bins are dropped. ‡: a different model, the fast gain shared within each band ("A fast gain shared across frequencies", `validate_shared_gain.json`), quoted at 180, 188 MHz, where the per-frequency fit collapses. Its error is the injection rms over 3 mocks of that model.

0 of them the fit is flagged diverged instead), and the ground, offset, transmitter and Sun terms grow to keep the level. The fit finds the second. Holding a_{tcal} near 4.5 with a prior of ± 0.2 in log brings it back above 0.50 at 3 of these 3 frequencies (0.8–1.6), and changes χ^2/dof by at most 0.24% and the held-out rms by at most 0.01 points. The prior is soft: its rows are not scaled by χ^2/dof, so these data can outweigh it (§7.5 uses a hard one). Where it does pull a_{tcal} back, the fit is no worse, so the data do not choose between the two solutions there. The sky is in those data. In the longest parked stretch (07-16 02:48–16:31 UTC, 30 m, 961 bins), raw power at these frequencies tracks the GSM column with correlations of 0.63–0.93, with no gain model at all; in a shorter 87.5 m stretch (07-17 16:29–18:50, 145 bins, where the sky changes less) 2 of the 3 exceed 0.5. In the long stretch the slope against the column is 3–1030 times the gain the collapsed fit puts there, where at the other frequencies the ratio is 0.19–2.2. In the collapsed solution the transmitter and Sun coefficients grow at least 15-fold. Then only the contrast between pointings ties the GSM's amplitude down, and at these frequencies that contrast is spoiled: the 4 windows with the transmitter on reach 9.8% residual, against a median of 0.5% in the others and 0.4% at 148–180 MHz. Dropping transmitter bins (3804 bins left) rescues 180, 188 MHz but turns 51, 82, 117, 125, 172, 227, 234 MHz into collapsed fits. χ^2/dof is 7.3–17.9 where the fit collapses, against 3.7–12.0 at the frequencies of the table. These frequencies are left out of the ground term. Earlier versions also found the fit collapsing at 51–74 MHz, with its 20-min gain following the sky there; that came with the rows on the ground. On rows at confirmed heights 59–74 MHz fit normally with the fast gain, so the daily term of §7.5 could be tested with it; this memo does not yet do so.

What the fast gain follows. Take the 20-min fit's log gain, average it over the frequencies of the table, and subtract a 3-h piecewise-linear fit: what is left varies by 1.9% rms over 110 knots, and the averages of two disjoint halves of those frequencies correlate at 0.96. `gain_drivers.py` projects every housekeeping stream in the correlator metadata onto the same knots, detrends it the same way and compares the two in two disjoint stretches: 07-15/16 at 30 m, and 07-17 at 30, 87.5 and 91 m (Fig. 20). Neither stretch shows the gain following the RF switch's temperature. The robust (Theil–Sen) slopes are -0.9 and -0.5%/K; a bootstrap over blocks of 6 knots, which respects the correlation between neighbouring knots, puts the least-squares slope between -2.6 and 0.3%/K on 07-15/16 and between -1.7 and 0.8%/K on 07-17 (95%), both including zero, and the temperature explains 0.07 and 0.01 of the variance. By fifths of the knots in temperature, the median fast gain runs from -0.3 to 1.0% in the four cooler fifths and is -0.6% in the warmest (-0.1 to 1.0 and -0.0% on 07-17). Earlier versions found the gain falling as the switch warmed; that relation came with the rows on the ground. The switch's fast temperature varies by 0.4 and 0.4 K rms. No other stream (the system current, the IMU, the motors, the pot, calibration and VNA cycles, the flag occupancy) correlates with the same sign at better than $|r| = 0.16$ in both stretches. Around the end of calibration cycles, VNA sweeps and antenna moves, the 3-h fit's residual moves by at most 0.7% (1.9 times its standard error), against up to 0.1% around random times. 93–99% of the fast gain's variance has no recorded cause.

The receiver's own gain. The calibration visits of `tcal@v0003` measure the gain after the RF switch without any sky model: the load's power over its temperature, and the noise source's excess over the load, which also carries the source's own drift. `validate_cal_gain.py` compares them with the fits (data-analysis commit 79e9960). They are dense only on 07-17, 04:11–16:22 UTC at 87.5 m, with the antenna mostly parked: 59 pairs 12 min apart. The 30 m era has too few to test, and nothing before the switch (antenna, balun, cable) is seen. Over that stretch (Fig. 19) the measured gain at 110–205 MHz varies by 0.8–1.3% rms by the load and 2.7–2.9% by the noise source, and follows the switch's thermistor (30.3–33.4 °C): the band's common mode changes by -1.2%/K by the load ($r = -0.93$) and -2.5%/K by the noise source ($r = -0.77$); the fast gain shows no such dependence ("What the fast gain follows"). After a running 3-h mean is subtracted, its correlation with the GSM column is -0.20 to 0.11 at every frequency: no 1–3 h structure follows the sky. Its common mode then varies by 0.5% rms (noise source), against 2.0% for the 20-min fit's gain (correlation 0.52). Over the same hours the 20-min fit's gain at the table's frequencies moves by 5.7–14.9% rms and follows the GSM column, with correlations up to 0.90 (141 MHz) and above 0.5 at 3 of the 10 frequencies of the table. In this parked stretch the fast gain is therefore not the receiver's gain after the switch. It is a gain change before the switch, or the fit absorbing a mismatch in the sky or beam model at a fixed pointing. The 30 m era has one usable interval. The raw correlator files of 07-16 (`validate_cal_thirty.py`; 39 files, SHA-256 recorded) hold 3 load visits. A fixed rule drops a visit that lies more than 10% from the visits' median at more than 2 of the 21 frequencies. It drops the first, at 01:14 UTC with only 22 rows, which fails at 6. Those frequencies all lie within 0.125 MHz of an integer MHz, and every frequency where it passes at least 0.19 MHz from one, as contamination by a comb with

1.000 MHz spacing would give. Memo 001 dates the onset of box-air's 1.000 MHz comb on 07-16 to 01:16:57, after this visit, so its source is not identified. Between the other two, 01:41–02:24 UTC (43 min, the antenna moving through the 07-16 raster), the load gain after the switch changes by −0.9 to 0.5% at 110–205 MHz and by at most 0.6% at 51–82 MHz. The 20-min fit's gain changes over the same minutes by −1.7 to 5.0% at 110–205 MHz and 3.4–15.4% at 51–82 MHz, while the GSM column rises 6.1–9.9%. Over that one interval, too, the fast gain is not the gain after the switch. With the antenna moving, it may there absorb a mismatch that follows the pointing rather than the sky, and one interval says nothing about 1–3 h structure or about the nights where the daily term of §7.5 is found. Fixing the gain to either calibration series, with one free level (Table 18), cannot test the daily term of §7.5 on 07-17. That stretch has none even with free 3-h knots (reflection with the real sky against a constant ground: −0.02 to 0.14 points). With the gain fixed, the held-out residual rises from 0.36–1.00% to 1.05–1.97%, and a shifted sky beats the real one at 6 (load) and 6 (noise source) of the 6 frequencies: the calibration gain is not accurate to the 1% this test needs. (The fixed-gain fits tie two gain knots one span apart. Earlier versions put them 10^6 s apart, which left the second without training bins, so `global_fit.py`'s held-out rule scored only one bin per frequency; those held-out values are replaced here.)

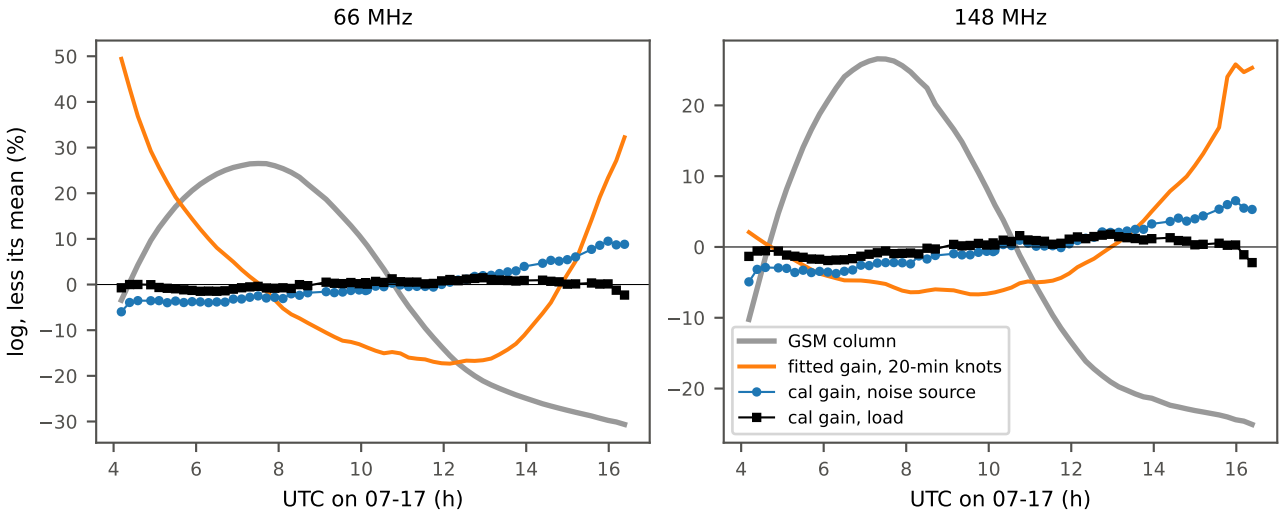


Figure 19: The receiver's gain after the RF switch from the calibration visits (`tcal@v0003`; load power over load temperature, and noise source minus load), the 20-min fit's gain at the same times, and the GSM column through the beam, each as a log less its mean, on 07-17 at 87.5 m (`validate_cal_gain.json`). At both frequencies the fitted gain moves far more than the measured gain.

	3-h knots			fixed, noise source			fixed, load		
MHz	const	real	shifted	const	real	shifted	const	real	shifted
62.5	0.85	+0.14	+0.13	1.83	−0.11	+0.65	1.11	+0.07	+0.36
66.4	0.46	+0.00	+0.01	1.88	−0.04	+0.82	1.07	+0.20	+0.52
70.3	0.56	+0.00	+0.00	1.97	−0.02	+0.77	1.15	+0.25	+0.47
74.2	0.50	+0.01	+0.00	1.86	−0.00	+0.77	1.09	+0.22	+0.52
78.1	0.38	−0.02	−0.01	1.82	−0.01	+0.71	1.05	+0.23	+0.48
82.0	0.53	−0.01	−0.01	1.84	+0.01	+0.66	1.06	+0.17	+0.40

Table 18: The daily term on 07-17 04:11–16:22 UTC (87.5 m, 628 bins; the phase-C model of §7.5 with $l = 0$; `validate_cal_gain.json`): held-out rms of a constant ground (%), and the improvement on it of reflection with the real sky and with the best of the ±6 h and 12 h shifted skies (percentage points). Gain free with 3-h knots and a soft amplitude anchor, or fixed to the calibration series from the noise source or the load, with one free level.

A fast gain shared across frequencies. The 20-min fit's fast gain is nearly the same at every frequency ("What the fast gain follows"), so one fast term could serve many frequencies, and a frequency could then no longer collapse by letting its own gain follow its sky. `validate_shared_gain.py` (data-analysis commit 79e9960) fits each frequency with a 3-h gain times one fast gain shared across the training frequencies: knots every 20 min, less what the 3-h gain can carry, with a gain and a tilt in frequency, or a quadratic in $\log\nu$. The shared gain and every training frequency's parameters are fitted jointly (Levenberg–Marquardt), until their steps are at most 9.8×10^{-7} in log gain (at most 714 iterations). An earlier version solved them in turn and did not converge, and its band comparison does not survive convergence. One selection rule is needed: a training frequency that collapses during the fit ($a_{\rm tcal} < 0.50$, or diverged) is dropped from the training set and the fit repeated, since otherwise it wanders along its own flat direction and drags the shared gain. The fits drop, trained on 117–203 MHz: none (linear), 117, 125, 180, 188, 195, 203 MHz (quadratic); trained on 51–203 MHz: 74, 156, 164, 172, 180, 188, 195, 203 MHz (linear), 51, 59 MHz (quadratic) and none (split, below). The injection refits apply the same rule. Of the 12, the ones that drop exactly what the real fit with the same training drops number 0 of 6 with 117–203 MHz training and 0 of 6 with 51–203 MHz. They drop 203 MHz in 3, 188 or 195 MHz in 0 and a low-band frequency in 0. So which frequencies collapse depends partly on structure in the real data that the mocks, built from the model's own prediction, do not carry.

Shared within each band, with its own gain and tilt below and above 100 MHz, the fast gain removes the collapse at 180, 188 MHz (2 of the 3; $a_{\rm tcal}$ =2.4–2.9). There its held-out rms is −0.11 to −0.07 points from the per-frequency fit's, and at the frequencies of Table 17 −0.20 to 0.15 points. Shared across both bands, the result depends on the training band. Trained on 51–203 MHz with a gain and a tilt, the fit drops 74, 156, 164, 172, 180, 188, 195, 203 MHz as they collapse, so above 150 MHz there is nothing left to compare; at 117–164 MHz $T_{\rm gnd}$ changes by up to 43% from the 117–203 MHz training (210 and 119 K at 148 MHz). A quadratic in $\log\nu$ is less stable: the two trainings differ by up to a factor 91.0. Against mocks built from the two quadratic fits, with three rolls of the real residuals and the same drop rule, that difference exceeds the injection scatter at 7 of 11 frequencies and is within it at 117, 148, 156, 180 MHz. So these data do not decide whether one fast term serves both bands, and the high-band ground term depends on that choice. At the frequencies the split fit rescues, mocks of the split fit itself (three rolls of the real residuals, the same drop rule) give its error. A value enters Table 17 only if no refit drops or collapses there and the injection rms is below 50% of the value. That admits 180, 188 MHz, with injection rms of 19.7–43.5 K, marked in the table as coming from this model. The two models differ: at 164 MHz the split fit gives −51 K where the table's per-frequency value is 250 K, so the marked rows continue the split fit's spectrum, not the table's.

7.7 Low-order sky corrections

The same regressions can add corrections to GSM: real spherical harmonics up to $l = 3$ in Galactic coordinates, whose fitted amplitudes over the GSM coefficient are corrections in GSM kelvin. Over an hour the sky turns only about 15° relative to the ground. So the same harmonics frozen to the ground (terrain, horizon or beam errors would look like that) are fitted too, and both are scored on held-out alternate 2-minute blocks (Fig. 21). The sky version beats the ground version at 14 of 25 frequencies in the raster, and in 2 of the 4 windows the corrections do not improve the held-out residual at most frequencies. Sky structure is not identified. With the harmonics in, $T_{\rm gnd}$ is no longer constrained either, since the lowest ones mimic the ground fraction. Telling sky from ground needs rotation at many sidereal times.

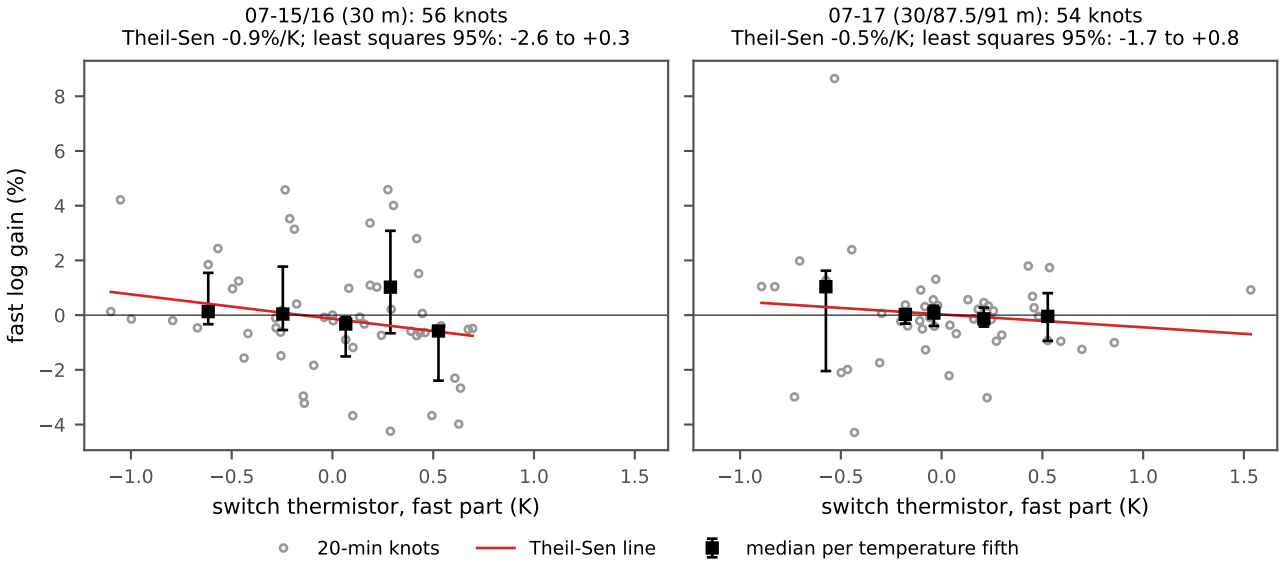


Figure 20: The fast log gain (20-min knots, mean over the frequencies of Table 17, less a 3-h piecewise-linear fit) against the RF switch's thermistor treated the same way, in two disjoint stretches of phase C (`validate_ground_term.json`). Grey: every knot; black: the median and interquartile range in each fifth of the temperature range; red: the Theil–Sen line, with the 95% block-bootstrap range of the least-squares slope above each panel. Neither stretch shows the gain following the temperature.

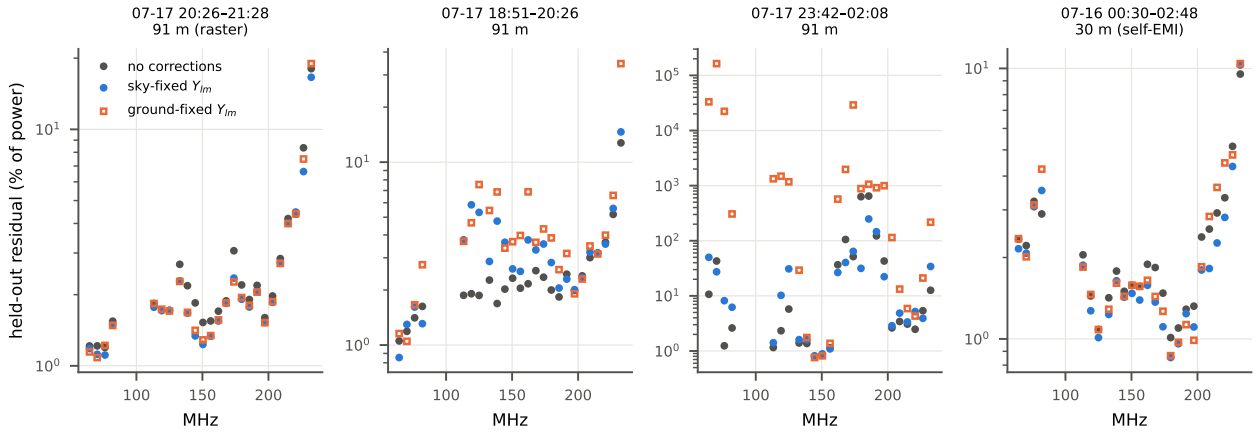


Figure 21: Held-out residual in each moving window, as a percentage of the power, without corrections to GSM (grey), with low-order sky-fixed corrections (blue) and with the same frozen to the ground (orange, open). The corrections do not reliably help: in 2 of the 4 windows (2 of the 3 at 91 m) the uncorrected model has the lowest held-out residual at most frequencies, and the sky-fixed corrections beat the ground-fixed ones at 14 of 25 frequencies in the raster. No sky structure is identified.

8 Open questions

- A horizon profile at 2 m (`horizon_profiles_v0005`) would bring back what is now left out because the antenna was on the ground: the 07-15 night, the ground half of the 07-18 night, and the geometry of the 07-17 sunset burst, including whether the terrain ended it (§4.1).
- What is $a \approx 2$: the kelvin scale, the beam, or both? Memo 004's noise-source temperature and memo 005's beam bear on it directly.
- The ground temperature needs calibrated data at more than one height. Extending `tcal` to the 30 m era would give the height lever of Table 1.
- box-gnd stands upright (like box-air at el = 0) with its box on the ground. Its position there is not yet solved. Before it can give a flux scale, a box-air × box-gnd cross-correlation, or a control for the Sun or for heating, its baseline, polarization and differential horizon need substantial work. It is not used here.
- Calibrating the 30 m-era crossings (07-16) would turn the raw-crossing fits of §6.7 into fluxes, and test whether the 07-17 excess is the Sun.
- What makes the ground term rise below 140 MHz? It is not specular reflection, horizon noise or localized sources, and not a fixed fraction of the sky (§7.4). The beam below the horizon, or the antenna's match changing with tilt, are left. A field measurement of S11 or a calibration cycle at several tilts would separate them.
- Separating sky corrections from ground-fixed structure needs rotation at many sidereal times.
- Is the low-frequency daily term over phase C sky, instrumental, or a gain the 3-h knots miss? On rows at confirmed heights the shifted skies fit nearly as well as the real one, and over phase C sidereal and solar time stay nearly in step (§7.5). Data weeks apart, or a recorded quantity that tracks it, would decide.
- What is the fast gain? Where the receiver's gain after the switch is measured it is not that, and 93–99% of it follows no recorded stream (§7.6). A gain change before the switch, or the fit absorbing sky or beam errors, is left; calibration cycles in the 30 m era, or a measurement of the antenna path's loss with temperature, would separate them. Until it is known, the gain needs knots every 20 min, and the frequencies where the fit then collapses have no ground term.
- The model is still far from the noise. Candidates are the beam (HFSS against the empirical beam), the sky prior, a time-variable ground, and the 10% GSM prior absorbing structure it cannot predict.

Revision history

2026-10-22 Heights from the mode table rebuilt on 2026-10-18 (memo 001), tested against box-air's lidar and IMU (§4.1), and pointing table v3. The 07-15 night and the 07-17 sunset were on the ground, not at 30 m, and there is no 2 m horizon profile, so every fit now uses only rows at 30, 87.5 or 91 m. Withdrawn: the 07-15 parked-night fit, the 07-17 dusk window, and the sunset burst's timing against the Sun going behind the ridge (the burst and its timing with the flare stand). Changed: the phase-C daily term no longer prefers the real sky over shifted skies; the low-band collapse of the fast-gain fit and the fast gain's relation to the RF switch's temperature both came with the ground rows and are gone; the impulsive events' rate is no longer the same with the Sun visible and behind the ridge, and one-variable controls (era, local hour, the Sun's height above the horizon) disagree, so their origin is open. The fixed-gain fits are now scored on their held-out bins, not on one.

2026-10-20 Added a fast gain shared across frequencies (§7.6), fitted jointly to convergence with collapsing training frequencies dropped by rule. It removed the collapse at most of the collapsing frequencies, and with a linear frequency dependence the high-band ground term changed little when the low band shared the gain. The ground-term table added its values where it rescued a frequency, marked, with injection errors.

2026-10-19 Moved every fit, product and validation to the extended terrain, DEM v0004 and `horizon_profiles_v0004`: `fit_v0013`–`v0016` and the `_v0004` products replace `fit_v0009`–`v0012` and the `_v0003` ones. No conclusion changes. The phase-C held-out comparisons move by a few hundredths

of a point; the largest other moves are the joint-fit floor on T_{gnd} and the night fit's unconstrained T_{gnd} values, both of which the text already treats as unconstrained. Added the 30 m era's one usable calibration interval (§7.6): there too the gain after the RF switch stays within 1% while the fitted fast gain moves several per cent.

2026-10-18 Added the ground term with a fast gain (§7.6): over phase C the gain changes within the hour, injections show 3-h knots bias T_{gnd} low, and with 20-min knots it is quoted at 110–205 MHz with injection errors. The fit collapsed at several low- and high-band frequencies, and the fast gain followed the RF switch's temperature. Where measured (07-17, 87.5 m), the receiver's gain after the switch moves about 1% and does not follow the sky, so the fast gain is not that gain; §7.5 notes what this leaves open for the daily term. Added the low-frequency daily term over phase C (§7.5): a slow additive term at 62–82 MHz that follows neither the pointing nor the recorded housekeeping. Diffuse reflection of the sky, and sky pickup, fit it over the whole phase. The sky origin and the reflection coefficient are withdrawn: the sidereal-time controls work only with the ground's level tied to R, and neither the preference nor R reproduces between 07-15 and 07-16. Faster gain knots (1 h, 20 min) take up the term into the gain, and a mock with a real term behaves the same way, so the term is stated as found with 3-h knots, with its additive nature open. Revised the 07-17 sunset at 30 m. The step in raw power there is mostly a solar burst during a C1.5 flare, ending as the disk goes behind the ridge before the elevation moves. Only the last 18% comes with the elevation step, which the earlier version blamed for all of it.

2026-10-16 Added the effect of extending the DEM 4 km south on the nominal horizon and on the predicted temperature (Section 2, Fig. 1), and the products that follow from it: DEM v0004 (6 km more to the south), `horizon_profiles_v0004` and its coverage check. No fit or product in this memo changed.

2026-10-07 Added the solar bursts, which identify solar emission in box-air; the raw context and raw-power fits of every terrain crossing; the local origin of the daytime impulsive events; the Sun's path through the beam and the raster test. The calibrated 07-17 excess is now read as larger than the Sun, not as the Sun. Added sky and ground recovery: the joint-frequency fit, the sky-calibrated parked nights, and the rotating-antenna regressions with their repeatable but frequency-dependent ground term. Traced the low-frequency excess: not specular reflection, horizon noise or localized sources, not a fixed fraction of the sky; the beam below the horizon or a tilt-dependent match remain, and need a field measurement.

2026-10-06 First version: the model, the degeneracy forecast, the night fit on `tcal@v0003` and `flags@v3`, and the first look for the Sun at the 07-17 terrain sunrise.

A Reproducing this memo

The products are regenerated, from `data-analysis/scripts/marjum-2026-07/ground_sky/` on data-analysis `main` (the products were built at 7093c01, cb0b00f and 79e9960, from the branch `ground-sky-pointing-v3`, since merged), with `EIGSEP_CAMPAIGN_ROOT` set to the campaign directory, by the commands below. They read the campaign's pointing table at v3 and take height eras from `curation/mode_table.jsonl` as rebuilt on 2026-10-18 (memo 001), by source file. Every product that depends on the antenna position reads the horizon profiles `curation/horizon_profiles_v0004` (geometry release `v0004`, memo 002) and traces the Sun on the DEM mosaic `derived/dem/v0004`, and every product that rotates a beam uses the empirical beam and mount offsets of `empirical_raster_v0013`. The burst cutouts depend on none of these; the binned spectra and the impulsive-event scan depend only on the pointing and the height eras.

- `python bin_spectra.py v0003` and `python bin_spectra.py v0004 -bin-s 60 -min-rows 10`; `python bin_raw.py v0002`;
- `python fit_ground_sky.py v0017 -binned v0003 -sun cut-flat`, `v0018 -binned v0003 -sun cut-terrain`, `v0019 -binned v0004 -sun cut-flat` and `v0020 -binned v0003 -sun model`;
- `python sun_events.py $EIGSEP_CAMPAIGN_ROOT/derived/ground_sky/fit_v0019 -out-dir $EIGSEP_CAMPAIGN_ROOT/derived/ground_sky/sun_events_v0005`;
- `python sun_bursts.py v0001`, `python impulsive_scan.py v0003`, `python sun_transitions.py v0005 -sun-events v0005 -fit v0019`, `python sun_profile.py v0005`, and `python raster_sun.py v0005 -beam hfss` then `-beam empirical`. The burst list they read is `curation/solar_bursts_v0001.json`.

- `python joint_fit.py v0005 -fit v0017`; `python sky_cal.py v0006 -beam hfss` and `-beam empirical`, each with `-fit-from 07-15=2026-07-15T05:12:00Z`; `python raster_sky.py v0005` with `-beam hfss -label raster`, `-beam empirical -label raster -reflect`, and for the other moving windows `-beam empirical` with the `-label`, `-start`, `-stop`, `-era` and `-row-step` recorded in each output (07-16 also `-self-emi`); and `python ground_excess.py v0005` with the `-split`, `-label` and window options recorded in each output.

The low-frequency daily term (§7.5) comes from `python validate_phase_c.py -data-analysis <checkout> -work <dir>` in this directory, and `validate_phase_c_fastgain.py` uses the same checkout. Every validation here takes a data-analysis checkout, checks that the commit it needs is an ancestor of it, and records the commit it ran at; the values below come from the commits named above, on the products listed there. The code was developed on separate branches and merged into main on 2026-10-21; re-running every validation from the merged code reproduced each number of the previous version of this memo. Commits from the branch `diffuse-ground` add the reflection, beam, pickup and housekeeping forms; the remaining options were added at c47da64 (`-ground-group`), fde5185 (`-ground-level-prior`), 6b10518 (`-t-range`), fc1dbb9 (`-night-only`, `-gain-segment-by-era`) and ae990d4 (`-t-exclude`). The 07-17 sunset burst comes from `python validate_sunset_burst.py -data-analysis <checkout>`, and the heights of §4.1 from `python validate_heights.py`, which reads every box-air raw file of 07-14 to 07-18 and records each one's SHA-256. Both record their inputs' checksums and the commit they ran at. The ground term with a fast gain (§7.6) comes from `python validate_ground_term.py -data-analysis <checkout> -work <dir>`. It runs `global_fit.py`, `inject_mock.py`, `hk_streams.py` and `gain_drivers.py` there, each pinned to at most five cores, and records the commit and the binned product's checksum. The receiver's own gain comes from `python validate_cal_gain.py -fit-checkout <dir> -cal-checkout <dir> -fit-work <dir> -cal-work <dir>`; it records both commits and the checksums of `raw_binned_v0002` and `tcal@v0003`. The 30 m-era load visits come from `python validate_cal_thirty.py -cal-checkout <dir> -fit-work <dir>`, run after `validate_ground_term.py` with the same work directory; it reads raw correlator files and records their SHA-256. The shared fast gain comes from `python validate_shared_gain.py -data-analysis <checkout> -fit-work <dir> -work <dir>`, run after `validate_ground_term.py` (whose per-frequency residuals its mocks use).

The commits they ran at are below; each product's `manifest.json` records them along with its inputs' checksums. This memo's figures, tables and numbers come from `python make_products.py` in this directory, then `pdflatex memo-008.tex` twice. The script checks the SHA-256 of every product file it reads.

input	SHA-256 (first 16)
derived/ground_sky/binned_v0003/binned.npz	3c7ea7bcb2d93564
derived/ground_sky/binned_v0003/manifest.json	7f572ebc9ecb11a9
derived/ground_sky/binned_v0004/binned.npz	3e0c808f2d85fdbd
derived/ground_sky/binned_v0004/manifest.json	ab2fa81fa056d0ca
derived/ground_sky/fit_v0017/fit.npz	8373922c014107e6
derived/ground_sky/fit_v0017/manifest.json	972bb895a8d190d2
derived/ground_sky/fit_v0018/fit.npz	03edf5105e47db49
derived/ground_sky/fit_v0018/manifest.json	f2a679f76a0bd67a
derived/ground_sky/fit_v0019/fit.npz	2388e9fdd163f16c
derived/ground_sky/fit_v0019/manifest.json	8a6149c801418967
derived/ground_sky/fit_v0020/fit.npz	8b005690c348d4d5
derived/ground_sky/fit_v0020/manifest.json	54f568d425726bd5
derived/ground_sky/sun_events_v0005/sun_events.json	add3464f0ccec53c
derived/ground_sky/sun_events_v0005/sun_events.npz	41a66688b6a9116b
derived/ground_sky/raw_binned_v0002/manifest.json	dd22d4cdac8b1ffd
derived/ground_sky/raw_binned_v0001/manifest.json	3ef90c83d3e3c0ad
derived/ground_sky/bursts_v0001/bursts.json	5e69db566e4dc2c0
derived/ground_sky/bursts_v0001/cutouts.npz	bca9c39857fd690e
derived/ground_sky/bursts_v0001/manifest.json	b9f52b6337f844e9
derived/ground_sky/impulsive_v0003/manifest.json	62abb32371b6beb6
derived/ground_sky/impulsive_v0003/events.jsonl	bf7a24d01cd6e88a
derived/ground_sky/transitions_v0005/transitions.json	a51e15abf7c61c46
derived/ground_sky/transitions_v0005/transitions.npz	66a873ab74b807b1
derived/ground_sky/transitions_v0005/manifest.json	cd6e2841665cc7bf

derived/ground_sky/sun_profile_v0005/sun_profile.npz	d0c77d080fa299ab
derived/ground_sky/sun_profile_v0005/manifest.json	ed401b83f1089946
derived/ground_sky/raster_sun_v0005/raster_hfss.json	3c807bdc47479973
derived/ground_sky/raster_sun_v0005/raster_empirical.json	95b33e0dea5590a8
derived/ground_sky/raster_sun_v0005/raster_empirical.npz	2a7e4533ea8d6774
derived/ground_sky/joint_v0005/joint_hfss.json	96fb2919ed202231
derived/ground_sky/skycal_v0006/skycal_hfss.json	d224b2907b89ab18
derived/ground_sky/skycal_v0006/skycal_empirical.json	725a00198c3863ef
derived/ground_sky/skycal_v0006/skycal_hfss.npz	ed9a3670831f98f1
derived/ground_sky/raster_sky_v0005/raster_hfss.json	b148f7b6ec022692
derived/ground_sky/ground_excess_v0005/raster_horizon_empirical.json	456223ad33bf4bff
derived/ground_sky/ground_excess_v0005/raster_horizon_noflags_empirical.json	e9ccf9b68f93d277
derived/ground_sky/ground_excess_v0005/raster_azimuth_empirical.json	ec1fc751e3e05d97
derived/ground_sky/ground_excess_v0005/w0718night91_horizon_empirical.json	96ca2d399cf9c22b
derived/ground_sky/ground_excess_v0005/w0718night91_azimuth_empirical.json	cfbdfd5443766292
derived/ground_sky/ground_excess_v0005/w0717eve91_azimuth_empirical.json	9500247d9bef44a0
derived/ground_sky/raster_sky_v0005/raster_empirical.json	5ec742f928b4f2f1
derived/ground_sky/raster_sky_v0005/w0717eve91_empirical.json	5331803cc82d1486
derived/ground_sky/raster_sky_v0005/w0718night91_empirical.json	bf621644831ee683
derived/ground_sky/raster_sky_v0005/w0716night_empirical.json	27902d611e41acf8
marjum-2026-07/curation/horizon_profiles_v0004.npz	1185ddbd39ff5a5f
data-analysis/scripts/marjum-2026-07/ground_sky/summary.json	44d2fc78f2df5b6b
memos/memo-008-marjum-2026-07-image-domain/inputs/solar_reference.json	e09895904559a8d0
validate_phase_c.json (this memo)	0d37834d36627318
validate_sunset_burst.json (this memo)	dd3c8eea662a3428
validate_sunset_burst.npz (this memo)	c198e636a7ec7140
validate_ground_term.json (this memo)	812ec91a0f0e019a
validate_cal_gain.json (this memo)	464e00790bff94ba
validate_cal_thirty.json (this memo)	27d29b1f87d52e17
validate_shared_gain.json (this memo)	2994ccf244568421
validate_phase_c_fastgain.json (this memo)	70775ef127aed830
validate_heights.json (this memo)	d3abe5647327ae9a
validate_heights.npz (this memo)	287f7d7899b770ba
validate_impulsive.json (this memo)	573460f8c0abf996
products made with data-analysis	347cecc, 7093c01, 79e9960, cb0b00f
products made with eigsep_base	840c10b
products made with eigsep_data	2b201d4, e941c51
products made with eigsep_sim	d050c83
phase-C fits: data-analysis ground-sky-pointing-v3	79e9960
fast-gain fits: data-analysis ground-sky-pointing-v3	79e9960
shared fast gain: data-analysis ground-sky-pointing-v3	79e9960
30 m receiver gain: data-analysis ground-sky-pointing-v3	79e9960
receiver-gain fits: data-analysis ground-sky-pointing-v3	79e9960
eigsep (memo) commit	70f05a2
data-analysis commit (degeneracy study file)	ac49dd3
generated	2026-10-23 18:49 UTC